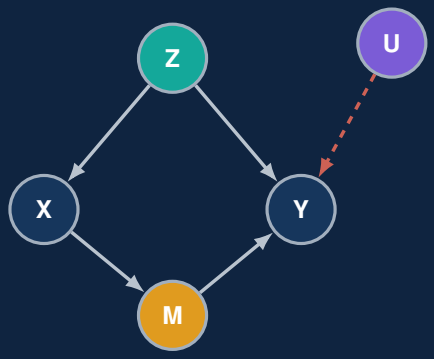


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 14

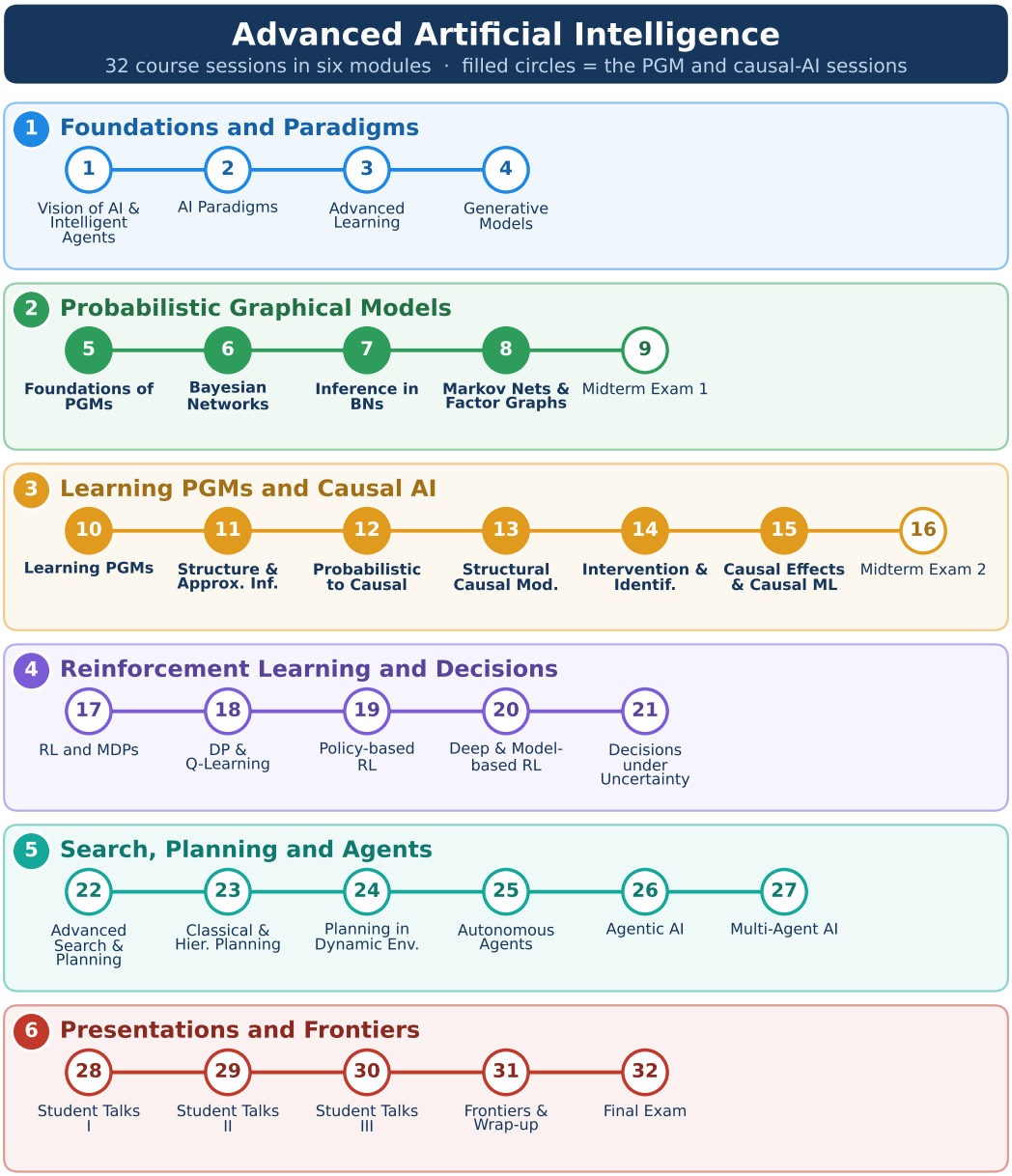
Intervention and Identification: Do-Calculus

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 14 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, a "you are here" course map, the core theory organized in parts, a dedicated part on mathematical statistics, fully worked numerical and analytical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s14_examples.py` for the numbers, `python/s14_figures.py` for the figures), and `questions/s14_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions.

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SESSION 14 ADVANCED ARTIFICIAL INTELLIGENCE

Intervention and Identification: Do-Calculus

The do-operator, do-calculus, back-door and front-door criteria, and identification

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Interventions
The do-operator, the identification problem, policies and soft interventions.

PART II
Back-door & Front-door
Adjustment, IPW, the kidney data, smoking–tar–cancer.

PART III
Do-Calculus & ID
Three rules, instruments, c-components, non-identifiability.

PART IV
Mathematical Statistics
Plug-in estimators, delta method, bootstrap, IV consistency, bounds.

PART V
AI Connections
RL actions, off-policy logs, tool-using agents, interacting agents.

LEARNING OBJECTIVES

- Define $P(y \mid \mathrm{do}(x))$, policy interventions $\mathrm{do}(\pi)$ and identifiability, and explain why identification depends on the graph, not on the sample size.
- State and prove the back-door adjustment formula and its IPW form; derive the front-door formula with the three rules of do-calculus.
- Go beyond both criteria with instruments, c-components and the ID algorithm; prove non-identifiability of the bow arc by constructing two SCMs.
- Derive the asymptotic distribution of the plug-in back-door estimator, compare it with the bootstrap, and analyse IV consistency and weak instruments.
- Turn a causal question into an estimand, with bounds and E-value sensitivity analysis as fall-backs.
- Recognise identification problems in RL, in tool-using LLM agents and in multi-agent systems.

Where we are. Session 12 showed that $P(y \mid x)$ can have the wrong sign as an answer to "what if we act?"; Session 13 built structural causal models (SCMs), in which $\mathrm{do}(X = x)$ is graph surgery computed *from the mechanisms*. This session (course session 14) drops that luxury: we know only a graph and observational data, and ask when $P(y \mid \mathrm{do}(x))$ is still computable. Session 15 estimates the identified quantities with machine learning; reinforcement learning (RL, course sessions 17–20), agentic AI (course session 26) and multi-agent AI (course session 27) need today's results whenever

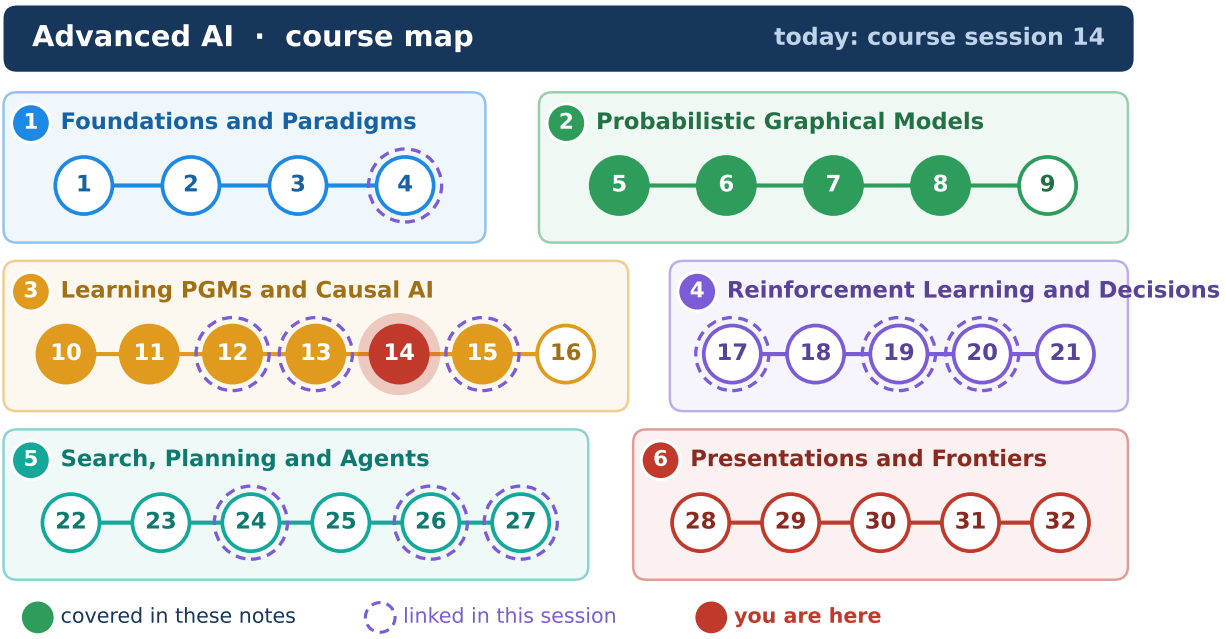


Figure 14.1. Where this session sits in the Advanced AI course. Dashed purple rings mark the course sessions linked today.

an agent learns about actions from data it did not generate.

OPENING MOTIVATION

Session 13's truncated factorization formula computes $P(y \mid \mathrm{do}(x))$ — *but only if we already have the full set of structural equations*, confounders and all. In practice we almost never observe every confounder: no dataset records "ability" directly, and no observational cancer study directly measures every genetic predisposition. The practitioner's real question is sharper: given only an observational dataset and a *graph* (which may include boxes for variables we know exist causally but cannot measure), can we still write $P(y \mid \mathrm{do}(x))$ as a formula using only observable quantities? This is the *identification* problem, and today's answer — the back-door criterion, the front-door criterion, and the complete rules of do-calculus — turns Session 12's confounding heuristics into checkable theorems.

Healthcare

Records show who got which treatment. Do they reveal recovery if *every* patient got A — or under a severity-based rule?

Economics and policy

Does schooling raise earnings when ability is unmeasured? An instrument may identify the effect where adjustment cannot.

LLM agents

Logs record when an old policy called a tool. Do they reveal what happens if the agent *always* calls it?

PART I

Interventions and the Identification Problem

What $\mathrm{do}(\cdot)$ means, what we may assume, and what we want to compute

PART I · INTERVENTIONS

14.1 The do-Operator Revisited

Let M be a Markovian SCM with causal DAG G over observed variables $\mathbf{V}$ (Session 13). The intervention $\mathrm{do}(X = x)$ replaces the mechanism of X by the constant x, producing the submodel M_x whose graph $G_{\overline{X}}$ has all arrows *into* X deleted. Its distribution is given by the truncated factorization

$$P(\mathbf{v} \mid \mathrm{do}(x)) = \prod_{i:\, V_i \notin X} P(v_i \mid \mathrm{pa}_i)\Big|_{X=x} \quad \text{for } \mathbf{v} \text{ consistent with } x, \qquad 0 \text{ otherwise.} \tag{14.1}$$

Throughout, we use the graph-surgery notation of Table 14.1.

Table 14.1. Mutilated graphs used by the back-door criterion and do-calculus.

Graph	Edges removed	Used to check
$G_{\overline{X}}$	all arrows *into* X	the world after $\mathrm{do}(x)$; Rule 1
$G_{\underline{X}}$	all arrows *out of* X	only back-door paths of X remain; Rule 2, back-door criterion
$G_{\overline{X}\underline{Z}}$	into X and out of Z	exchanging $\mathrm{do}(z)$ and z (Rule 2)
$G_{\overline{X}\,\overline{Z(W)}}$	into X, and into the Z-nodes that are not ancestors of W	deleting $\mathrm{do}(z)$ (Rule 3)

For the confounded triangle $Z \to X$, $Z \to Y$, $X \to Y$, Eq. (14.1) with $\mathbf{V} = \{Z, X, Y\}$ gives $P(z, y \mid \mathrm{do}(x)) = P(z)P(y \mid x, z)$, hence $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z)$, while the law of total probability gives $P(y \mid x) = \sum_z P(y \mid x, z)P(z \mid x)$. Seeing reweights the strata by $P(z \mid x)$; doing keeps $P(z)$. Both are computable — *because* Z was observed. This session is about what happens when it is not.

PART I · INTERVENTIONS

14.2 The Identification Problem

Definition 14.1 (Identifiability)**.** A causal effect $P(y \mid \mathrm{do}(x))$ is *identifiable* from a causal graph G if it can be expressed as a function of the observational distribution $P(\mathbf{v})$ over the observed variables alone — computable without knowledge of the specific structural equations or unobserved-variable distributions, using only G's structure and observational data.

Formally: for every pair of SCMs M_1, M_2 that induce the graph G and satisfy $P_{M_1}(\mathbf{v}) = P_{M_2}(\mathbf{v}) > 0$, we require $P_{M_1}(y \mid \mathrm{do}(x)) = P_{M_2}(y \mid \mathrm{do}(x))$. Three consequences follow. (i) Identification concerns the *population* ("could infinite data reveal it?"); more data never rescues a non-identified effect, which is not estimated with large variance but undefined by the data. (ii) To *disprove* it, exhibit two models that agree on $P(\mathbf{v})$ and disagree on $P(y \mid \mathrm{do}(x))$ (Section 14.9). (iii) Positivity matters: a formula dividing by $P(x \mid z)$ is meaningless where $X = x$ never occurs.

PART I · INTERVENTIONS

14.3 Interventions Beyond do(x): Policies and Soft Interventions

Real decisions are rarely atomic: a doctor treats by severity, a recommender randomizes, an RL agent follows a stochastic policy. Pearl (2009, Ch. 4) treats such *policy* interventions by the same surgery (Fig. 14.2).

Definition 14.2 (Conditional, stochastic and soft interventions)**.** Let Z be a set of non-descendants of X. A *conditional intervention* $\mathrm{do}(X = g(Z))$ replaces the mechanism of X by the deterministic rule $x = g(z)$. A *stochastic policy* $\mathrm{do}(\pi)$ replaces it by a fresh random draw $X \sim \pi(\cdot \mid Z)$, independent of all other exogenous noise given Z. More generally, a *soft* (mechanism-changing) intervention replaces $P(x \mid \mathrm{pa}_X)$ by any new kernel, possibly with new parents. Atomic interventions are the special case $\pi(x' \mid z) = \mathbb{1}(x' = x)$.

Proposition 14.1 (Value of a policy)**.** *If Z consists of non-descendants of X and satisfies the back-door criterion (Definition 14.4) relative to (X, Y), then for every policy $\pi(x \mid z)$*

$$P\big(y \mid \mathrm{do}(\pi)\big) = \sum_{z} \sum_{x} P(y \mid x, z)\, \pi(x \mid z)\, P(z). \tag{14.2}$$

Proof. Let M_π be the SCM in which $X := h(Z, \varepsilon)$ with $P(h(z,\varepsilon) = x) = \pi(x \mid z)$ and ε independent of all original noises. Then $P_\pi(y) = \sum_{z,x} P_\pi(z) P_\pi(x \mid z) P_\pi(y \mid x, z)$. (a) The mechanisms of the non-descendants of X and their noises are untouched, so $P_\pi(z) = P(z)$. (b) $P_\pi(x \mid z) = \pi(x \mid z)$ by construction. (c) Given $Z = z$, the event $\{X = x\}$ in M_π is determined by ε, which is independent of everything else; conditioning on it is therefore the same as setting $X = x$, and $P_\pi(y \mid x, z) = P(y \mid \mathrm{do}(x), z)$. Finally $P(y \mid \mathrm{do}(x), z) = P(y \mid x, z)$ by Rule 2 of do-calculus, because Z blocks every back-door path (Section 14.6; this is the second step of the proof in Section 14.19). □

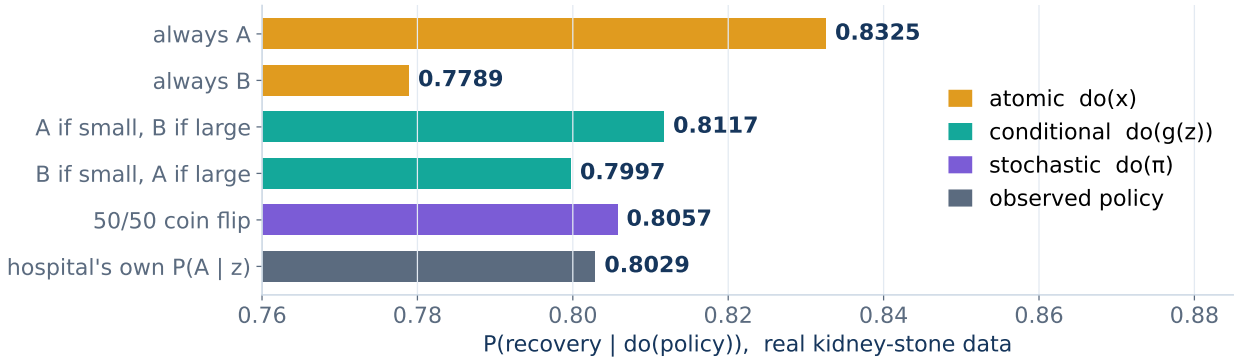


Figure 14.2. Values of atomic, conditional and stochastic interventions on the real kidney-stone data (Example 14.1). The hospital's own policy reproduces the observed recovery rate $562/700$, as it must.

Worked Example 14.1 (Policies on the kidney-stone data)**.** Take the real Charig et al. (1986) counts of Session 12 (Table 14.2), with $X \in \{A, B\}$ the treatment, $Y = 1$ for recovery and Z = stone size, a valid back-door set (Worked Example 14.2). Write $\mu_{t,z} = P(Y{=}1 \mid X{=}t, Z{=}z)$;

Table 14.2. The real kidney-stone data (Charig et al. 1986; Session 12): recovered/treated patients. Treatment A = open surgery, B = percutaneous nephrolithotomy (PCNL); Z = stone size.

Stone size Z	Treatment A	Treatment B	All patients
small	$81/87 = 0.9310$	$234/270 = 0.8667$	357 $(P(\text{small}) = 357/700 = 0.51)$
large	$192/263 = 0.7300$	$55/80 = 0.6875$	343 $(P(\text{large}) = 343/700 = 0.49)$
pooled	$273/350 = 0.7800$	$289/350 = 0.8257$	$562/700 = 0.8029$ recovered

from the table $\mu_{A,s} = 0.9310$, $\mu_{B,s} = 0.8667$, $\mu_{A,\ell} = 0.7300$, $\mu_{B,\ell} = 0.6875$ (s = small, ℓ = large), and $P(s) = 0.51$, $P(\ell) = 0.49$. For a policy that gives A with probability $\pi(A \mid z)$, Eq. (14.2) reads $P(Y{=}1 \mid \text{do}(\pi)) = \sum_z P(z)\big[\pi(A \mid z)\mu_{A,z} + (1 - \pi(A \mid z))\mu_{B,z}\big]$, so any treatment rule can be evaluated without running it:

$$\begin{aligned}
\text{always A:} \quad & 0.51(0.9310) + 0.49(0.7300) = 0.4748 + 0.3577 = 0.8325, \\
\text{always B:} \quad & 0.51(0.8667) + 0.49(0.6875) = 0.4420 + 0.3369 = 0.7789, \\
\text{A if small, B if large:} \quad & 0.51(0.9310) + 0.49(0.6875) = 0.4748 + 0.3369 = 0.8117, \\
\text{B if small, A if large:} \quad & 0.51(0.8667) + 0.49(0.7300) = 0.4420 + 0.3577 = 0.7997, \\
\text{fair coin, } \pi(A \mid z) = 0.5: \quad & \tfrac{1}{2}(0.8325) + \tfrac{1}{2}(0.7789) = 0.8057.
\end{aligned}$$

The hospital's actual policy is $\pi(A \mid s) = P(A \mid \text{small}) = 87/357 = 0.2437$ and $\pi(A \mid \ell) = P(A \mid \text{large}) = 263/343 = 0.7668$. Its within-stratum recovery rates are $0.2437(0.9310) + 0.7563(0.8667) = 0.8824$ (small) and $0.7668(0.7300) + 0.2332(0.6875) = 0.7201$ (large), so its value is $0.51(0.8824) + 0.49(0.7201) = 0.8029 = 562/700$. This is no coincidence: with $\pi(x \mid z) = P(x \mid z)$, Eq. (14.2) becomes $\sum_{z,x} P(y \mid x, z)P(x \mid z)P(z) = P(y)$, the observed recovery rate — a useful sanity check. Since A is better in both strata, no rule beats "always A". (All values are printed to four decimals; each is computed from the unrounded fractions.)

PYTHON LAB · `s14_examples.py` · `ex_policy_kidney()`

```
atomic do(A) 0.8325; do(B) 0.7789; A if small/B if large 0.8117; B if
small/A if large 0.7997
coin flip 0.8057; observed policy P(A|z) 0.8029 = 562/700
observed policy, stratum small:  0.2437(0.9310) + 0.7563(0.8667) = 0.8824;
large:  0.7668(0.7300) + 0.2332(0.6875) = 0.7201
```

PART II

The Back-door and Front-door Criteria

Two graphical certificates that turn $P(y \mid \text{do}(x))$ into observational quantities

PART II · ADJUSTMENT

14.4 The Back-door Criterion

Definition 14.3 (Back-door path)**.** A path from X to Y is a *back-door path* if its first edge points *into* X, i.e. it has the form $X \leftarrow \cdots Y$. (Paths whose first edge is $X \rightarrow \cdots$ are not back-door paths, whether or not they are directed.)

Definition 14.4 (Back-door criterion)**.** A set of variables Z satisfies the back-door criterion relative to (X, Y) if: (i) no node in Z is a descendant of X, and (ii) Z blocks every back-door path between X and Y (in the sense of d-separation, Session 6).

Condition (ii) is a single d-separation statement: Z blocks all back-door paths iff $X \perp\!\!\!\perp_d Y \mid Z$ in $G_{\underline{X}}$, since removing the arrows out of X leaves exactly the back-door paths. Checking the criterion is therefore as cheap as one Bayes-ball run (linear in the size of the graph, Session 6).

Theorem 14.1 (Back-door adjustment)**.** *If Z satisfies the back-door criterion relative to (X, Y), then $P(y \mid \mathrm{do}(x))$ is identifiable and equals*

$$P\big(y \mid \mathrm{do}(x)\big) = \sum_{z} P(y \mid x, z)\, P(z). \tag{14.3}$$

Proof for $Z = \mathrm{pa}(X)$, from the truncated factorization. Write $T = \mathrm{pa}(X)$ and let $R = \mathbf{V} \backslash (\{X, Y\} \cup T)$. On configurations with $X = x$, Eq. (14.1) and the Markov factorization $P(\mathbf{v}) = P(x \mid t) \prod_{i \neq X} P(v_i \mid \mathrm{pa}_i)$ give, wherever $P(x \mid t) > 0$,

$$P(\mathbf{v} \mid \mathrm{do}(x)) = \frac{P(\mathbf{v})}{P(x \mid t)}.$$

Summing over r and t: $P(y \mid \mathrm{do}(x)) = \sum_t \sum_r P(x, y, t, r)/P(x \mid t) = \sum_t P(x, y, t)/P(x \mid t) = \sum_t P(y \mid x, t) P(t)$, which is Eq. (14.3) with $Z = T$. The parents of X always satisfy the back-door criterion, so this is the "adjustment for direct causes" special case. For a general back-door set Z (which may exclude unobserved parents of X) the result follows from do-calculus; the derivation is in Section 14.19. □

This is exactly the stratify-and-reweight calculation Session 12's real-world investigators performed by hand on the Berkeley and kidney-stone data — now justified as the unique correct formula whenever Z satisfies a checkable graphical condition, rather than a statistical instinct to "control for confounders."

Worked Example 14.2 (Formally resolving the kidney-stone paradox)**.** In the SCM $X := g(Z, U_X)$, $Y := h(X, Z, U_Y)$ from Session 13, stone severity Z satisfies the back-door criterion for (X, Y): it is not a descendant of treatment X (severity is determined before treatment is chosen), and it blocks the only back-door path $X \leftarrow Z \rightarrow Y$. Applying Theorem 14.1 to the *real* data from Session 12 (Table 14.2, Fig. 14.3): $P(Z{=}\text{small}) = (87 + 270)/700 = 357/700 = 0.510$, $P(Z{=}\text{large}) = (263 + 80)/700 = 343/700 = 0.490$, and the stratum recovery rates are

$81/87 = 0.9310$, $192/263 = 0.7300$ for A and $234/270 = 0.8667$, $55/80 = 0.6875$ for B. So

$$P(\text{recover} \mid \text{do}(A)) = 0.9310(0.510) + 0.7300(0.490) = 0.4748 + 0.3577 = 0.8325,$$
$$P(\text{recover} \mid \text{do}(B)) = 0.8667(0.510) + 0.6875(0.490) = 0.4420 + 0.3369 = 0.7789.$$

Treatment A's interventional recovery probability (0.8325) now correctly exceeds Treatment B's (0.7789), matching what both size-stratified subgroups already showed individually in Session 12. The naive pooled rates $P(\text{recover} \mid A) = 273/350 = 0.7800$ and $P(\text{recover} \mid B) = 289/350 = 0.8257$ weight the strata by $P(z \mid t)$ instead of $P(z)$: $P(\text{large} \mid A) = 263/350 = 0.7514$ but $P(\text{large} \mid B) = 80/350 = 0.2286$, so A is charged with the hard cases. Theorem 14.1 is precisely the formula that makes re-weighting by $P(z)$ the mathematically *correct* way to combine the two strata. The *average treatment effect* (ATE) of A versus B is $P(\text{recover} \mid \text{do}(A)) - P(\text{recover} \mid \text{do}(B)) = 0.83255 - 0.77887 = +0.0537$ (difference of the unrounded values), against the naive $0.7800 - 0.8257 = -0.0457$.

PYTHON LAB · s14_examples.py · ex_kidney_backdoor()

```
P(Z=small) = 357/700 = 0.510; naive P(rec|A) = 273/350 = 0.7800, P(rec|B) =
289/350 = 0.8257
P(rec|do(A)) = 0.4748 + 0.3577 = 0.8325, P(rec|do(B)) = 0.4420 + 0.3369 =
0.7789
P(large|A) = 0.7514, P(large|B) = 0.2286; naive A-B = -0.0457, adjusted ATE
= 0.83255 - 0.77887 = +0.0537
```

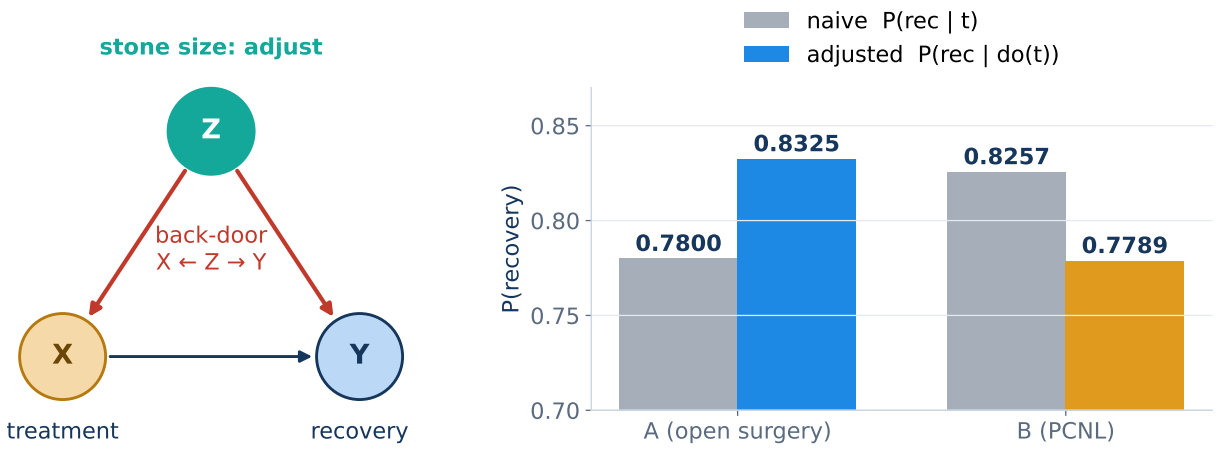


Figure 14.3. Left: stone severity Z satisfies the back-door criterion. Right: naive $P(\text{rec} \mid t)$ versus adjusted $P(\text{rec} \mid \text{do}(t))$ on the real 1986 data (Table 14.2, Worked Example 14.2); adjustment restores the correct ranking of treatments.

Worked Example 14.3 (Checking the back-door criterion carefully)**.** In the kidney-stone graph (Fig. 14.3), verify condition (i) of the back-door criterion: is stone severity Z a descendant of treatment X? *Solution:* No — severity is a biological fact determined before any treatment decision, so $Z \notin \text{de}(X)$, and condition (i) holds. This matters because adjusting for a *post-treatment* variable (e.g., a side-effect of treatment that also affects recovery) can introduce exactly the collider bias warned about in Sessions 6 and 12 — the back-door criterion's descendant restriction is precisely what rules this mistake out.

14.4.1 Several back-door sets: which to choose?

Worked Example 14.4 (All back-door sets of a six-node graph). *(Numerical, graph computation.)* In the six-node DAG of Fig. 14.4 ($Z_1 \to Z_3 \leftarrow Z_2$, $Z_1 \to X$, $Z_3 \to X$, $Z_3 \to Y$, $Z_2 \to Y$, $X \to W \to Y$), enumerate all $2^4 = 16$ subsets of $\{Z_1, Z_2, Z_3, W\}$ and test both conditions with networkx d-separation in $G_{\underline{X}}$. There are four back-door paths: (a) $X \leftarrow Z_3 \to Y$, (b) $X \leftarrow Z_3 \leftarrow Z_2 \to Y$, (c) $X \leftarrow Z_1 \to Z_3 \to Y$, and (d) $X \leftarrow Z_1 \to Z_3 \leftarrow Z_2 \to Y$, on which Z_3 is a collider. *Result:* exactly three sets are valid, $\{Z_1, Z_3\}$, $\{Z_2, Z_3\}$ and $\{Z_1, Z_2, Z_3\}$; the first two are minimal. Conditioning on Z_3 is necessary, since it is the only observed non-collider on (a). But $\{Z_3\}$ alone fails: conditioning on the collider Z_3 opens path (d), which must then be re-blocked by Z_1 or Z_2 (this is *M-bias*, named after the M shape $Z_1 \to Z_3 \leftarrow Z_2$). Any set containing the mediator W violates condition (i).

PYTHON LAB · s14_examples.py · ex_backdoor_sets()

```
back-door paths:  X-Z1-Z3-Y, X-Z1-Z3-Z2-Y, X-Z3-Y, X-Z3-Z2-Y
valid sets:  [('Z1', 'Z3'), ('Z2', 'Z3'), ('Z1', 'Z2', 'Z3')]
minimal valid sets:  [('Z1', 'Z3'), ('Z2', 'Z3')]
```

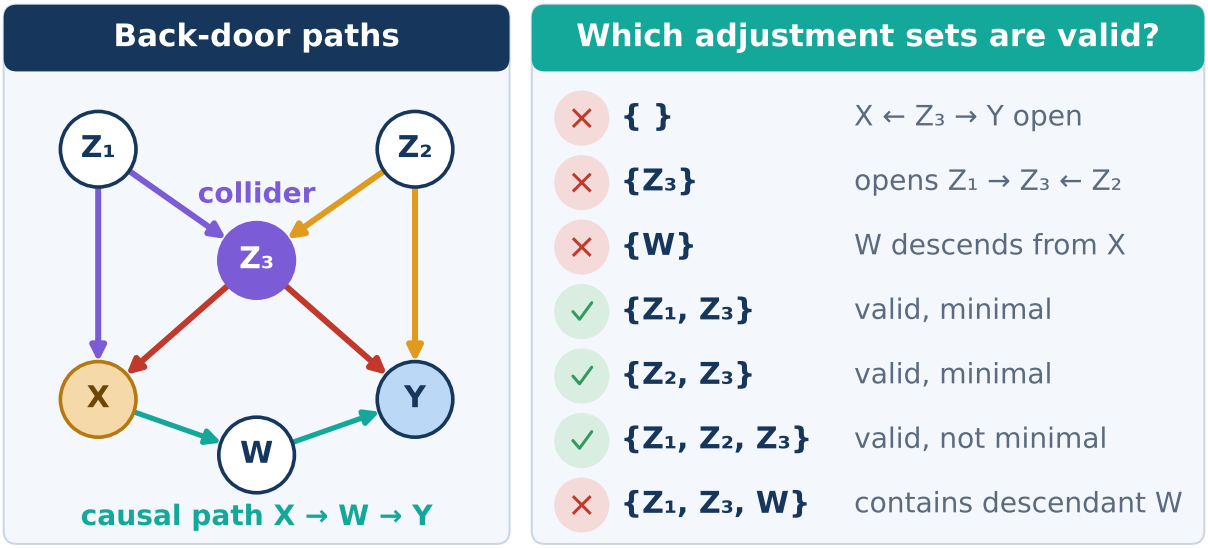


Figure 14.4. Back-door paths of the six-node DAG of Worked Example 14.4 (red: $X \leftarrow Z_3 \to Y$; purple then amber: the path $X \leftarrow Z_1 \to Z_3 \leftarrow Z_2 \to Y$ through the collider Z_3; the other two back-door paths combine red edges with one purple or amber edge) and the verdict for candidate adjustment sets.

All valid sets give the *same* population value $P(y \mid \mathrm{do}(x))$; they differ in finite-sample variance (Session 15).

14.4.2 Adjustment as weighting: the IPW representation

Inverse-probability weighting (IPW) rewrites the adjustment formula as a weighted average over units, with the *propensity* $P(x \mid z)$ in the denominator.

Proposition 14.2 (IPW form of the adjustment formula). *Let Z satisfy the back-door criterion*

and suppose positivity, $P(x \mid z) > 0$ whenever $P(z) > 0$. Then

$$P(y \mid \mathrm{do}(x)) = \mathbb{E}\left[\frac{\mathbb{1}(X = x)\mathbb{1}(Y = y)}{P(x \mid Z)}\right], \qquad \mathbb{E}[Y \mid \mathrm{do}(x)] = \mathbb{E}\left[\frac{\mathbb{1}(X = x)\, Y}{P(x \mid Z)}\right]. \tag{14.4}$$

Proof. Condition on Z: $\mathbb{E}\big[\mathbb{1}(X{=}x)\mathbb{1}(Y{=}y)/P(x \mid Z)\,\big|\,Z = z\big] = P(X{=}x, Y{=}y \mid z)/P(x \mid z) = P(y \mid x, z)$. Taking the expectation over Z gives $\sum_z P(y \mid x, z)P(z)$, which is Eq. (14.3). The second identity follows by summing $y\,P(y \mid \mathrm{do}(x))$ over y. □

Adjustment *models the outcome*; IPW *models the treatment*, reweighting units by $1/P(x \mid z)$ into a pseudo-population in which $X \perp\!\!\!\perp Z$ — a simulated randomized trial. Session 15 combines both (doubly robust estimation).

Worked Example 14.5 (IPW on the kidney data)**.** *(Numerical, data of Table 14.2.)* The propensities of A are $e(\text{small}) = \hat{P}(A \mid \text{small}) = 87/357 = 0.2437$ and $e(\text{large}) = 263/343 = 0.7668$, so the weights are $1/e = 357/87 = 4.1034$ for A–small, $343/263 = 1.3042$ for A–large, $1/(1-e) = 357/270 = 1.3222$ for B–small and $343/80 = 4.2875$ for B–large: the rare combinations (A on small stones, B on large stones) count about four times. Only recovered patients ($Y = 1$) contribute to $\frac{1}{700}\sum_i \mathbb{1}(X_i = t)Y_i/\hat{P}(t \mid Z_i)$:

$$\text{A:}\quad \big(81 \times 4.1034 + 192 \times 1.3042\big)/700 = (332.38 + 250.40)/700 = 0.8325,$$
$$\text{B:}\quad \big(234 \times 1.3222 + 55 \times 4.2875\big)/700 = (309.40 + 235.81)/700 = 0.7789,$$

exactly the adjustment values. This is an algebraic identity for discrete Z with estimated propensities: in stratum z each of the R_{tz} recovered patients with treatment t gets weight N_z/N_{tz} (N_z patients in the stratum, N_{tz} of them treated with t), so the IPW average is $\sum_z \frac{N_z}{N} \cdot \frac{R_{tz}}{N_{tz}}$, which is the adjustment formula with $\hat{P}(z) = N_z/N$ and $\hat{P}(Y{=}1 \mid t, z) = R_{tz}/N_{tz}$.

PYTHON LAB · s14_examples.py · ex_ipw_kidney()

```
IPW (Horvitz-Thompson, empirical propensities):  P(rec|do(A)) = 0.8325,
P(rec|do(B)) = 0.7789
weights:  A-small 357/87 = 4.1034, A-large 343/263 = 1.3042, B-small 357/270
= 1.3222, B-large 343/80 = 4.2875
A: (332.38 + 250.40)/700 = 0.8325; B: (309.40 + 235.81)/700 = 0.7789
```

14.4.3 Linear models: adjustment is multiple regression

In a linear SEM, $\mathbb{E}[Y \mid \mathrm{do}(x), z]$ is linear in (x, z), so Eq. (14.3) says: regress Y on X *and* a back-door set Z and read off the coefficient of X. Adding the wrong variable is dangerous:

Worked Example 14.6 (Over-control: adjusting for a mediator)**.** *(Analytical, with a simulation check.)* Consider the linear SEM with a mediator W: $Z = U_Z$, $X = 0.8Z + U_X$, $W = X + U_W$, $Y = 1.5W + 2Z + U_Y$, where U_Z, U_X, U_W, U_Y are independent $N(0,1)$ and there is no direct $X \to Y$ arrow. The total effect is $\partial\mathbb{E}[Y \mid \mathrm{do}(x)]/\partial x = 1 \times 1.5 = 1.5$ (the product of the coefficients along $X \to W \to Y$), and $\{Z\}$ is a back-door set. *Regression on (X, Z):* substituting W, $Y = 1.5X + 2Z + (1.5U_W + U_Y)$ with an error independent of (X, Z), so the population coefficients are (1.5, 2) — correct. *Regression on (X, Z, W):*

$Y = 0 \cdot X + 2Z + 1.5W + U_Y$ with U_Y independent of (X, Z, W), so these are already the population normal equations and the coefficients are (0, 2, 1.5). Adjusting for the descendant W removes the *entire* effect — the coefficient of X is 0, not 1.5. This is condition (i) of the back-door criterion at work. Simulation (`ex_overcontrol()`, $n = 2 \times 10^5$, seed 15): 1.500 with $\{Z\}$; −0.002 with $\{Z, W\}$.

PYTHON LAB · `s14_examples.py` · `ex_overcontrol()`

```
simulation n=200000, seed=15:  Y   X + Z : coef on X = 1.500
Y   X + Z + W : coef on X = -0.002, on Z = 2.000, on W = 1.502
```

PART II · ADJUSTMENT

14.5 The Front-Door Criterion: When Adjustment Is Impossible but Identification Still Succeeds

The back-door criterion requires *measuring* enough of the confounding structure to block every back-door path. What if the confounder itself is unmeasurable?

Worked Example 14.7 (Smoking, tar, and cancer)**.** Consider X = smoking, Y = lung cancer, with a hypothesized unmeasured genetic factor U that predisposes people both to smoke and (independently) to develop cancer — a classic historical objection to the smoking-cancer causal claim. Suppose additionally that smoking's effect on cancer is entirely mediated by tar deposits in the lungs, M: $X \to M \to Y$, with U affecting X and Y directly but *not* M (Fig. 14.5).

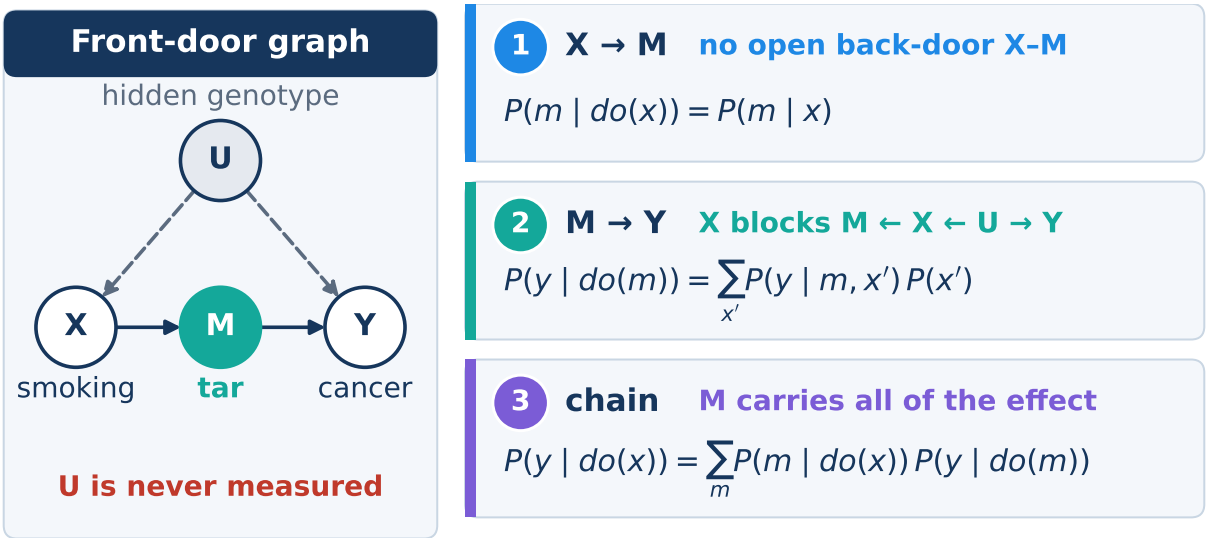


Figure 14.5. The front-door configuration: $X \to Y$ is confounded by unobserved U, but the mediating path through M is unconfounded in both legs. Right: the three do-calculus steps of Theorem 14.2's derivation.

Definition 14.5 (Front-door criterion)**.** A set of variables M satisfies the front-door criterion relative to (X, Y) if: (i) M intercepts all directed paths from X to Y; (ii) there is no unblocked back-door path from X to M; (iii) every back-door path from M to Y is blocked by X.

Theorem 14.2 (Front-door adjustment)**.** *If M satisfies the front-door criterion relative to (X, Y) and $P(x, m) > 0$, then $P(y \mid \mathrm{do}(x))$ is identifiable and equals*

$$P(y \mid \mathrm{do}(x)) = \sum_{m} P(m \mid x) \sum_{x'} P(y \mid x', m)\, P(x'). \tag{14.5}$$

Derivation by do-calculus (graph of Fig. 14.5). 1 *Effect of X on M.* In $G_{\underline{X}}$ the only path between X and M is $X \leftarrow U \rightarrow Y \leftarrow M$, blocked at the collider Y; so $M \perp\!\!\!\perp_d X$ in $G_{\underline{X}}$ and Rule 2 gives $P(m \mid \mathrm{do}(x)) = P(m \mid x)$.

2 *Effect of M on Y.* Condition on X: $P(y \mid \mathrm{do}(m)) = \sum_{x'} P(y \mid \mathrm{do}(m), x') P(x' \mid \mathrm{do}(m))$. In $G_{\overline{M}}$ the arrow $X \rightarrow M$ is cut and the only X–M path $X \leftarrow U \rightarrow Y \leftarrow M$ is blocked at Y, so Rule 3 gives $P(x' \mid \mathrm{do}(m)) = P(x')$. In $G_{\underline{M}}$ the remaining M–Y path $M \leftarrow X \leftarrow U \rightarrow Y$ is blocked by conditioning on X, so Rule 2 gives $P(y \mid \mathrm{do}(m), x') = P(y \mid m, x')$. Hence $P(y \mid \mathrm{do}(m)) = \sum_{x'} P(y \mid m, x') P(x')$.

3 *Chain the legs.* $P(y \mid \mathrm{do}(x)) = \sum_{m} P(y \mid \mathrm{do}(x), m) P(m \mid \mathrm{do}(x))$. In $G_{\overline{X}\underline{M}}$ ($U \rightarrow X$ and $M \rightarrow Y$ cut) X is attached only to M, so $Y \perp\!\!\!\perp_d M \mid X$ and Rule 2 gives $P(y \mid \mathrm{do}(x), m) = P(y \mid \mathrm{do}(x), \mathrm{do}(m))$; in $G_{\overline{X}\,\overline{M}}$ the node X is isolated, so Rule 3 gives $P(y \mid \mathrm{do}(x), \mathrm{do}(m)) = P(y \mid \mathrm{do}(m))$. Substituting steps 1 and 2 yields Eq. (14.5). □

INTUITION

The formula estimates the effect in two unconfounded legs and chains them: $P(m \mid x)$ is the (unconfounded, by condition ii) effect of smoking on tar; the inner sum $\sum_{x'} P(y \mid x', m) P(x')$ estimates the (unconfounded once we further adjust for X, by condition iii) effect of tar on cancer, holding the confounding influence of U on Y fixed by averaging over X's marginal distribution rather than conditioning on the specific x we started with. The genuinely surprising part is that the unmeasured, un-adjustable confounder U between X and Y never needs to be measured at all — it is defeated by routing around it through a mediator both of whose legs happen to be individually unconfounded.

Worked Example 14.8 (Front-door numerically)**.** Suppose (illustrative numbers) $P(M{=}1 \mid X{=}1) = 0.8$, $P(M{=}1 \mid X{=}0) = 0.2$ (smoking strongly increases tar), $P(X{=}1) = 0.4$ in the population, and $P(Y{=}1 \mid X{=}1, M{=}1) = 0.5$, $P(Y{=}1 \mid X{=}0, M{=}1) = 0.5$, $P(Y{=}1 \mid X{=}1, M{=}0) = 0.1$, $P(Y{=}1 \mid X{=}0, M{=}0) = 0.1$ (in these illustrative numbers cancer risk depends only on tar: $P(y \mid x, m)$ does not vary with x). Then $\sum_{x'} P(Y{=}1 \mid x', M{=}1) P(x') = 0.5(0.4) + 0.5(0.6) = 0.5$ and similarly $\sum_{x'} P(Y{=}1 \mid x', M{=}0) P(x') = 0.1$. So $P(Y{=}1 \mid \mathrm{do}(X{=}1)) = P(M{=}1 \mid X{=}1)(0.5) + P(M{=}0 \mid X{=}1)(0.1) = 0.8(0.5) + 0.2(0.1) = 0.42$, versus $P(Y{=}1 \mid \mathrm{do}(X{=}0)) = 0.2(0.5) + 0.8(0.1) = 0.18$ — a causal effect of smoking on cancer of $0.42 - 0.18 = 0.24$, computed without measuring U. Note, however, that here the naive contrast is the same: $P(Y{=}1 \mid X{=}1) = 0.8(0.5) + 0.2(0.1) = 0.42$ and $P(Y{=}1 \mid X{=}0) = 0.2(0.5) + 0.8(0.1) = 0.18$. Because $P(y \mid x, m)$ does not depend on x, these particular numbers carry no trace of confounding by U (given M, X tells us nothing about Y, hence nothing about U's effect on Y). The next example has genuine confounding.

PYTHON LAB · `s14_examples.py` · `ex_frontdoor_numeric()`

```
inner sums:  m=1 -> 0.50, m=0 -> 0.10; P(Y=1|do(X=1)) = 0.42, P(Y=1|do(X=0))
= 0.18, effect = 0.24
```

Worked Example 14.9 (Front-door estimation with a hidden confounder)**.** *(Numerical, simulation; hypothetical SCM.)* Let $U \sim \text{Bern}(0.5)$ be unobserved, $X \sim \text{Bern}(0.2 + 0.6U)$, $M \sim \text{Bern}(0.1 + 0.7X)$ and $Y \sim \text{Bern}(0.1 + 0.5M + 0.3U)$; U does not enter M, so the front-door criterion holds. *Truth:* $P(Y{=}1 \mid \text{do}(x)) = 0.1 + 0.5\,P(M{=}1 \mid x) + 0.3\,P(U{=}1)$, i.e. $0.1 + 0.5(0.8) + 0.3(0.5) = 0.65$ for $x = 1$ and $0.1 + 0.5(0.1) + 0.3(0.5) = 0.30$ for $x = 0$, an effect of 0.35. *Naive:* by Bayes, $P(U{=}1 \mid X{=}1) = 0.5(0.8)/(0.5(0.8) + 0.5(0.2)) = 0.8$ and $P(U{=}1 \mid X{=}0) = 0.2$, so $\mathbb{E}[Y \mid X{=}1] = 0.1 + 0.5(0.8) + 0.3(0.8) = 0.74$ and $\mathbb{E}[Y \mid X{=}0] = 0.1 + 0.5(0.1) + 0.3(0.2) = 0.21$: the naive contrast 0.53 overstates the effect by half. *Front-door:* plugging the empirical frequencies of $n = 50\,000$ simulated (X, M, Y) triples (seed 14) into Eq. (14.5) gives 0.652 and 0.301, an effect of 0.351 — the truth, without ever seeing U.

PYTHON LAB · s14_examples.py · ex_frontdoor_sim()

```
truth:  P(Y=1|do(1)) = 0.650, P(Y=1|do(0)) = 0.300, ATE = 0.350; population
naive 0.74 - 0.21 = 0.53
frontdoor (n=50000, seed=14):  do(1) = 0.652, do(0) = 0.301, ATE = 0.351
```

PART III

Do-Calculus and Identification

Three rules, instruments, c-components, and the limits of identification

PART III · DO-CALCULUS

14.6 Do-Calculus: The Complete Rules

Back-door and front-door are the two most commonly invoked special cases of a fully general calculus for manipulating do(·) expressions, built entirely from d-separation applied to mutilated graphs. Write $G_{\overline{X}}$ for G with X's incoming edges removed (Session 13's mutilation) and $G_{\underline{X}}$ for G with X's outgoing edges removed.

Theorem 14.3 (Rules of do-calculus, Pearl 1995)**.** *For disjoint variable sets X, Y, Z, W:*

Rule 1 (insertion/deletion of observations): $P(y \mid \text{do}(x), z, w) = P(y \mid \text{do}(x), w)$ *if* $(Y \perp\!\!\!\perp_d Z \mid X, W)$ *in* $G_{\overline{X}}$.

Rule 2 (action/observation exchange): $P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), z, w)$ *if* $(Y \perp\!\!\!\perp_d Z \mid X, W)$ *in* $G_{\overline{X}\underline{Z}}$.

Rule 3 (insertion/deletion of actions): $P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), w)$ *if* $(Y \perp\!\!\!\perp_d Z \mid X, W)$ *in* $G_{\overline{X}\,\overline{Z(W)}}$, *where* $Z(W)$ *excludes any* Z*-nodes that are ancestors of* W *in* $G_{\overline{X}}$.

These three rules are sound and complete*: a causal effect is identifiable if and only if it can be reduced to a purely observational expression by some finite sequence of applications of Rules 1–3 (together with ordinary probability axioms).*

We state completeness without proof (it is a substantial graph-theoretic result; Shpitser & Pearl,

2006, give an algorithm that always finds the reduction when one exists). What matters for practice is the intuition behind each rule: every one of them is exactly Session 6's d-separation criterion, applied not to the original graph but to a graph mutilated by the actions already performed — do-calculus adds nothing conceptually new to d-separation, it only asks us to check separation in the right (possibly surgically modified) graph for each manipulation we want to justify. Both Theorem 14.1 and Theorem 14.2 can be derived as short sequences of these three rules; we did this for the front-door case above and do it for the back-door case in the appendix.

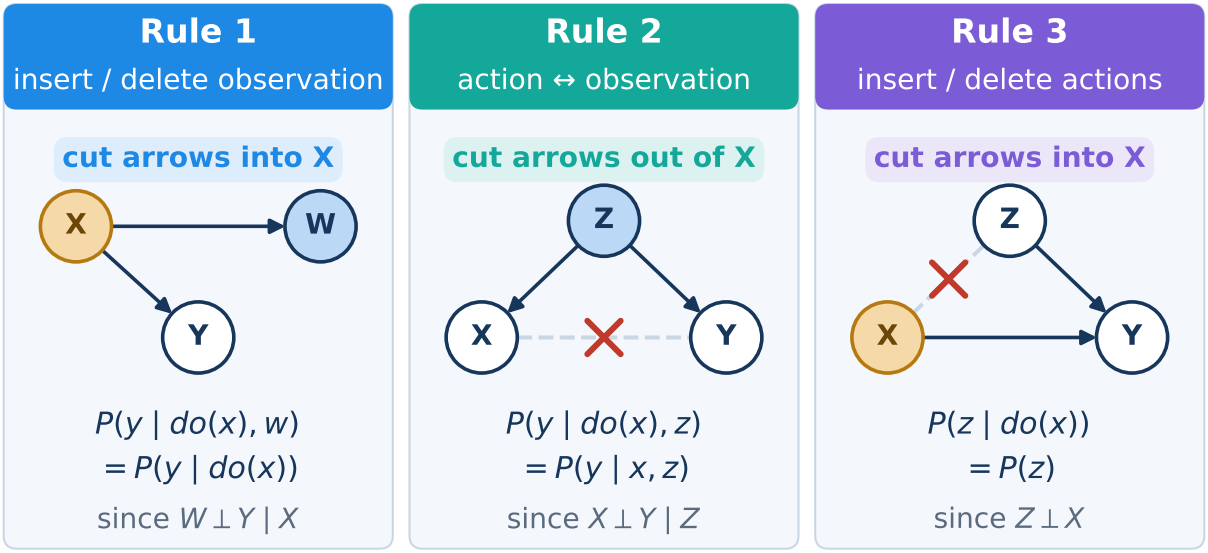


Figure 14.6. The three rules as cards, each with the surgery used in its minimal example ($\perp$ denotes d-separation in the cut graph). Rule 1 drops an irrelevant observation (Worked Example 14.10); Rule 2 turns $\mathrm{do}(x)$ into x once the back-door path is blocked by Z; Rule 3 deletes an action that cannot reach the target. In the Rule 2 and Rule 3 cards the action being exchanged or deleted is X, which plays the role of Z in Theorem 14.3; hence the arrows cut are those out of, respectively into, X.

Why the rules are sound (sketch). Give each intervened variable Z a "switch" parent F_Z (idle or a forced value), so that $P(y \mid \mathrm{do}(z), \cdot) = P(y \mid F_Z{=}z, \cdot)$; each rule is then an ordinary d-separation statement in the augmented DAG (Rule 2: F_Z is irrelevant once Z is observed; Rule 3: F_Z is irrelevant altogether), and soundness follows from that of d-separation and the causal Markov condition.

Worked Example 14.10 (An explicit application of Rule 1)**.** Let $X \to Y$ and also $X \to W$, with no edge between W and Y at all (W is an unrelated side-effect of X). Show $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x))$ using Rule 1.

Solution. Rule 1 requires $Y \perp\!\!\!\perp_d W \mid X$ in $G_{\overline{X}}$ (delete edges into X; here X has none, so $G_{\overline{X}} = G$). In G, the only path between W and Y is $W \leftarrow X \to Y$, a fork with fork-node X; since X is in the conditioning set, this path is blocked, and there is no other path (by assumption W, Y share no direct edge), so $Y \perp\!\!\!\perp_d W \mid X$ holds. Rule 1 therefore licenses dropping w from the conditioning set: $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x))$. Practically, this says an irrelevant post-treatment side effect of X — one that does not itself influence Y — can always be safely ignored when computing Y's response to the intervention, without needing to measure or adjust for it at all.

PART III · DO-CALCULUS

14.7 Instrumental Variables

Back-door needs the confounders, front-door needs a clean mediator. A third classical route uses a variable that moves X *as if at random*.

Definition 14.6 (Instrument)**.** Z is an *instrumental variable* for the effect of X on Y if (a) *relevance*: Z and X are dependent; (b) *exclusion*: every directed path from Z to Y passes through X; (c) *exogeneity*: Z shares no unblocked back-door path with Y (e.g. $Z \perp\!\!\!\perp U$ for the hidden confounder U of X and Y).

Without further assumptions an instrument does *not* point-identify $P(y \mid \mathrm{do}(x))$ (the bow arc between X and Y remains; only bounds follow). Linearity (below) or monotonicity of X in Z restores identification; under monotonicity the identified quantity is the *local average treatment effect* (LATE) for the "compliers", the units whose treatment is moved by the instrument (Imbens & Angrist 1994).

Proposition 14.3 (IV identification in the linear model)**.** *Let* $X = \alpha Z + \delta U + \varepsilon_X$ *and* $Y = \beta X + \gamma U + \varepsilon_Y$ *with* $Z, U, \varepsilon_X, \varepsilon_Y$ *mutually independent and first-stage coefficient* $\alpha \neq 0$. *Then*

$$\beta = \frac{\mathrm{Cov}(Z,Y)}{\mathrm{Cov}(Z,X)}, \qquad \textit{and for binary } Z: \quad \beta = \frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[X \mid Z{=}1] - \mathbb{E}[X \mid Z{=}0]} \quad \textit{(Wald)}. \tag{14.6}$$

The OLS slope instead tends to $\beta + \gamma\delta\mathrm{Var}(U)/\mathrm{Var}(X)$.

Proof. By bilinearity and independence, $\mathrm{Cov}(Z,Y) = \beta\mathrm{Cov}(Z,X) + \gamma\mathrm{Cov}(Z,U) + \mathrm{Cov}(Z,\varepsilon_Y) = \beta\mathrm{Cov}(Z,X)$ and $\mathrm{Cov}(Z,X) = \alpha\mathrm{Var}(Z) \neq 0$. For binary Z with $p = P(Z{=}1)$, $\mathrm{Cov}(Z,V) = p(1-p)\big(\mathbb{E}[V \mid Z{=}1] - \mathbb{E}[V \mid Z{=}0]\big)$ for any V, and the factor $p(1-p)$ cancels. For OLS, $\mathrm{Cov}(X,Y) = \beta\mathrm{Var}X + \gamma\mathrm{Cov}(X,U) = \beta\mathrm{Var}X + \gamma\delta\mathrm{Var}U$; divide by $\mathrm{Var}X$. □

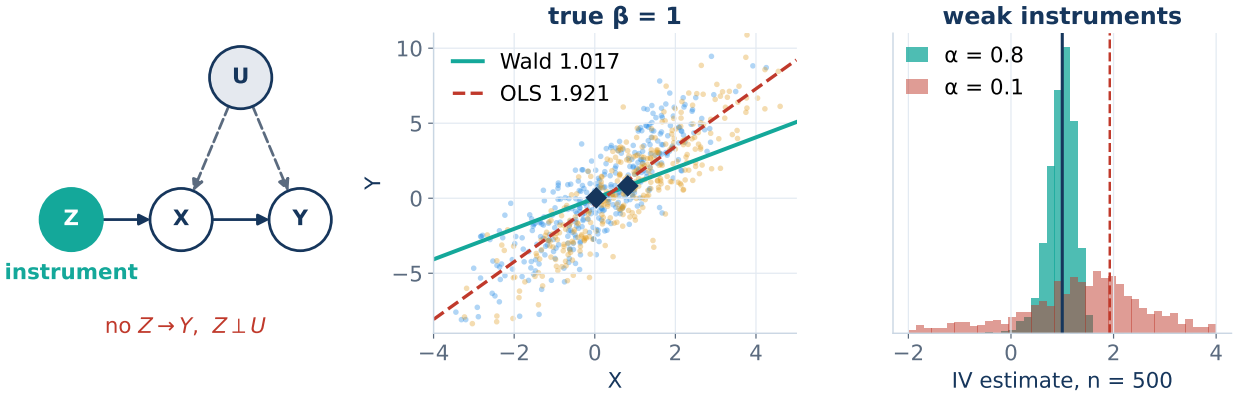


Figure 14.7. Instrumental variables (hypothetical linear model, $\beta = 1$). Left: the IV graph. Middle: one simulated sample of Worked Example 14.11 ($n = 5000$, 700 points shown); the two Z-group means (diamonds) lie on the Wald line of slope $1.017 \approx \beta$, while OLS (1.921) is pulled towards $\beta + \gamma\delta\mathrm{Var}U/\mathrm{Var}X = 1.926$. Right: sampling distributions of the Wald estimator over 2000 samples of size $n = 500$ for a strong ($\alpha = 0.8$) and a weak ($\alpha = 0.1$) instrument; solid line $\beta = 1$, dashed line the OLS limit 1.926; estimates outside $[-2, 4]$ are not shown.

Worked Example 14.11 (Wald estimator versus OLS). *(Numerical, simulation.)* Hypothetical model of Proposition 14.3 with $\alpha = 0.8$, $\delta = 1$, $\beta = 1$, $\gamma = 2$: $Z \sim \text{Bern}(0.5)$, $U, \varepsilon_X, \varepsilon_Y \sim N(0,1)$, all independent, $X = 0.8Z + U + \varepsilon_X$, $Y = X + 2U + \varepsilon_Y$. Here $\text{Var}Z = 0.5(0.5) = 0.25$ and $\text{Var}X = 0.8^2(0.25) + 1 + 1 = 2.16$, so OLS tends to $1 + 2(1)(1)/2.16 = 1 + 0.926 = 1.926$. One sample of $n = 5000$ (`ex_iv_sim()`, seed 14): OLS slope 1.921; group means $(\bar{x}, \bar{y}) = (0.0375, 0.0318)$ for $Z = 0$ and $(0.8181, 0.8252)$ for $Z = 1$, so the Wald ratio is $(0.8252 - 0.0318)/(0.8181 - 0.0375) = 0.7934/0.7805 = 1.017$, close to $\beta = 1$.

PYTHON LAB · s14_examples.py · ex_iv_sim()

```
Var X = 0.8^2 (0.25) + 1 + 1 = 2.16; plim OLS = 1 + 2/2.16 = 1.926
n=5000, alpha=0.8, seed=14:  OLS slope = 1.921 (plim 1.926); Wald = 1.017
group means:  Z=0 (x=0.0375, y=0.0318), Z=1 (x=0.8181, y=0.8252); Wald =
0.7934/0.7805 = 1.017
```

PART III · DO-CALCULUS

14.8 c-Components and the ID Algorithm in Outline

When neither criterion applies, the complete answer is algorithmic. Represent each hidden common cause by a bidirected edge, giving Session 13's *acyclic directed mixed graph* (ADMG).

Definition 14.7 (c-component). Two observed variables are in the same *c-component* (confounded component) if they are connected by a path consisting only of bidirected edges. The c-components $S_1, \dots, S_k$ partition $\mathbf{V}$. For $S \subseteq \mathbf{V}$ define $Q[S](\mathbf{v}) = P(s \mid \text{do}(\mathbf{v} \setminus s))$, the joint law of S when everything else is set by intervention.

Theorem 14.4 (c-component factorization; Tian & Pearl 2002). $P(\mathbf{v}) = \prod_{j=1}^{k} Q[S_j]$, *and every factor is identified: with a topological order* $V_1 < \cdots < V_n$,

$$Q[S_j] = \prod_{i:\, V_i \in S_j} P(v_i \mid v_1, \dots, v_{i-1}).$$

Moreover, if $A \subseteq C$ *is ancestral in the subgraph induced by* C, *then* $Q[A] = \sum_{c \setminus a} Q[C]$.

The ID algorithm (Shpitser & Pearl 2006), in outline. To compute $P(y \mid \text{do}(x))$: (1) let D be the set of ancestors of Y (including Y) in the graph $G[\mathbf{V} \setminus X]$ obtained by deleting X; then $P(y \mid \text{do}(x)) = \sum_{d \setminus y} Q[D]$. (2) Split D into its c-components $D_1, \dots, D_m$ in $G[D]$, so $Q[D] = \prod_j Q[D_j]$. (3) Compute each $Q[D_j]$ from the identified $Q[S]$ of the c-component $S \supseteq D_j$ of G, by repeatedly marginalizing ancestral subsets and re-factorizing into c-components (Theorem 14.4). The algorithm fails exactly when some step meets a *hedge* — two nested c-forests, the generalisation of the bow arc — and then the effect is provably not identifiable.

Worked Example 14.12 (The front-door formula from c-components). *(Analytical.)* In the front-door graph the c-components are $\{X, Y\}$ (via $X \leftrightarrow Y$) and $\{M\}$. With order $X < M < Y$: $Q[\{M\}] = P(m \mid x)$ and $Q[\{X, Y\}] = P(x)P(y \mid x, m)$. Step 1: deleting X leaves $M \to Y$,

so the ancestors of Y are $D = \{M, Y\}$ and $P(y \mid \mathrm{do}(x)) = \sum_m Q[\{M, Y\}]$. Step 2: in $G[D]$ there is no bidirected edge, so $Q[\{M, Y\}] = Q[\{M\}]\, Q[\{Y\}]$. Step 3: $\{Y\}$ is ancestral within $G[\{X, Y\}]$ (which has no directed edge), so $Q[\{Y\}] = \sum_{x'} Q[\{X, Y\}] = \sum_{x'} P(x')P(y \mid x', m)$ (here m is a fixed argument, since M is intervened on in $Q[\{X, Y\}]$). Together: Eq. (14.5). In the bow-arc graph, by contrast, $D = \{Y\}$ lies in the c-component $\{X, Y\}$ but $\{Y\}$ is not ancestral there (the arrow $X \to Y$ is inside it): the hedge.

PART III · DO-CALCULUS

14.9 Non-Identifiability: When No Formula Exists

Worked Example 14.13 (The bow-arc graph)**.** Let U be an unobserved common cause of both X and Y, with also a direct edge $X \to Y$, and no measured mediator at all (Fig. 14.8). No set of *observed* variables can satisfy the back-door criterion (the only back-door path $X \leftarrow U \to Y$ passes through the unmeasured U, and there are no other measured variables to substitute), and the front-door criterion requires a mediator M intercepting the $X \to Y$ edge, which does not exist here either.

Theorem 14.5 (Non-identifiability of the bow-arc graph)**.** *For the graph in Fig. 14.8, no application of do-calculus (Theorem 14.3) can reduce $P(y \mid \mathrm{do}(x))$ to an expression in $P(x, y)$ alone; there exist two distinct SCMs sharing this graph and inducing the identical observational distribution $P(x, y)$ but different interventional distributions $P(y \mid \mathrm{do}(x))$.*

Proof. It suffices to construct the two models (then, by soundness of do-calculus, no derivation exists). Fix any strictly positive $P(x, y)$ on $\{0, 1\}^2$. *Model A:* $U = (U_X, U_Y)$ with independent components, $X := U_X$ with $P(U_X{=}1) = P(X{=}1)$, and $Y := \mathbb{1}\{U_Y < P(Y{=}1 \mid X)\}$ with $U_Y \sim \mathrm{Unif}(0, 1)$. It reproduces $P(x, y)$, and since $U_Y \perp\!\!\!\perp X$, $P_A(Y{=}1 \mid \mathrm{do}(x)) = P(Y{=}1 \mid x)$. *Model B:* $U = (U_1, U_2)$ drawn jointly from $P(x, y)$, $X := U_1$, $Y := U_2$. It also reproduces $P(x, y)$, but Y does not listen to X, so $P_B(Y{=}1 \mid \mathrm{do}(x)) = P(Y{=}1)$ for both x. Both models are compatible with the bow arc (an SCM may leave an arrow of the graph unused). The two interventional distributions coincide only if $P(Y{=}1 \mid x) = P(Y{=}1)$, i.e. if $X \perp\!\!\!\perp Y$; whenever the data show any dependence, they differ. □

Worked Example 14.14 (Two SCMs, one dataset)**.** *(Numerical.)* Take $P(X{=}1) = 0.5$, $P(Y{=}1 \mid X{=}1) = 0.8$, $P(Y{=}1 \mid X{=}0) = 0.3$, i.e. $P(x, y) = P(x)P(y \mid x) = (0.5 \cdot 0.7,\ 0.5 \cdot 0.3,\ 0.5 \cdot 0.2,\ 0.5 \cdot 0.8) = (0.35, 0.15, 0.10, 0.40)$ for $(x, y) = (0, 0), (0, 1), (1, 0), (1, 1)$. Model A gives $P(Y{=}1 \mid \mathrm{do}(x)) = P(Y{=}1 \mid x)$: 0.80 and 0.30, an ATE of $0.80 - 0.30 = 0.50$. Model B gives $P(Y{=}1 \mid \mathrm{do}(x)) = P(Y{=}1) = 0.15 + 0.40 = 0.55$ for both x, an ATE of 0. The data cannot tell a strong effect from no effect at all.

PYTHON LAB · `s14_examples.py` · `ex_nonident()`

```
Model A: P(Y=1|do(1)) = 0.80, P(Y=1|do(0)) = 0.30, ATE = 0.50
Model B: P(Y=1|do(1)) = 0.55, P(Y=1|do(0)) = 0.55, ATE = 0.00
bounds:  P(Y=1|do(1)) in [0.40, 0.90], P(Y=1|do(0)) in [0.15, 0.65]; ATE in
[-0.25, 0.75]
```

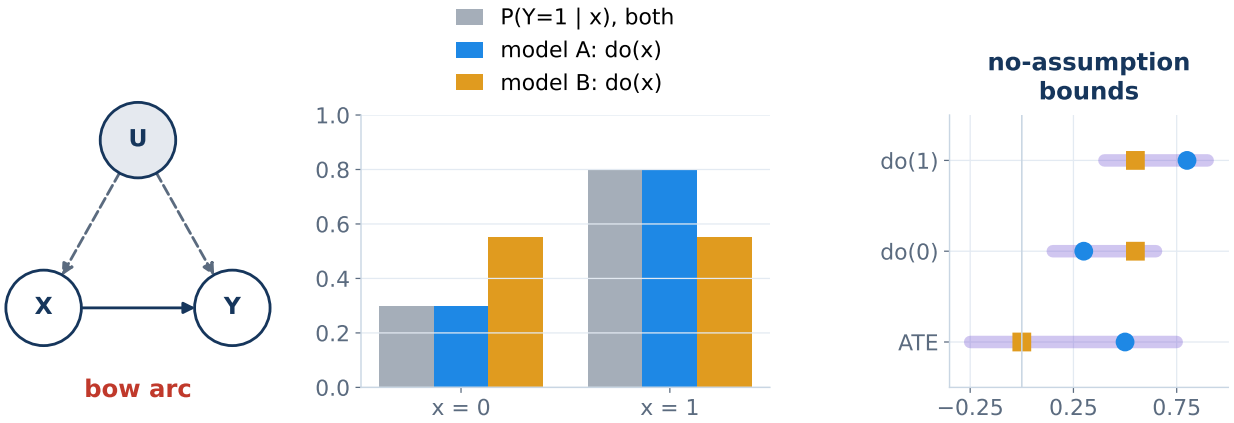


Figure 14.8. A structure for which $P(y \mid \mathrm{do}(x))$ is provably not identifiable from $P(x, y)$ alone. Middle: two SCMs with identical $P(Y{=}1 \mid x)$ (grey) but different interventional probabilities. Right: the no-assumption bounds of Proposition 14.5 (purple) contain both models (circle: A, square: B).

This is not a failure of cleverness — it is a mathematical impossibility, and recognizing it (rather than reporting a confidently wrong number) is itself the valuable skill this session teaches. When a causal graph implies non-identifiability, the only paths forward are: collect more data (measure a proxy for U, breaking the bow), run a randomized experiment (Session 15's benchmark case, which sidesteps identification entirely by design), or bound the effect rather than pin it down exactly (partial identification, Proposition 14.5).

PART III · DO-CALCULUS

14.10 From Question to Estimand: A Practical Recipe

1 *Question:* write the target as $P(y \mid \mathrm{do}(x))$, an ATE or a policy value $P(y \mid \mathrm{do}(\pi))$. 2 *Graph:* draw the causal DAG with domain experts, unmeasured variables included; every missing arrow is an assumption. 3 *Identify* (Table 14.3): back-door, front-door, an instrument, then the ID algorithm; if all fail, randomize, measure a proxy, bound, or change the question. 4 *Estimate* the functional (Part IV, Session 15). 5 *Stress-test* testable implications (Session 13), positivity and hidden confounding.

Table 14.3. Identification strategies compared.

Strategy	Must observe	Key assumption	Estimand
Back-door	a set Z blocking all back-door paths	no hidden confounder outside Z; positivity	$\sum_z P(y \mid x, z)P(z)$ or IPW
Front-door	mediators M	M captures the whole effect; both legs unconfounded	$\sum_m P(m \mid x) \sum_{x'} P(y \mid x', m)P(x')$
Instrument	Z affecting X	exclusion, exogeneity, plus linearity or monotonicity	Wald ratio / LATE
ID algorithm	any	the full ADMG is correct	products and sums of $Q[S]$
None	—	—	bounds, sensitivity, experiment

Remark 14.1 (Sensitivity analysis and the E-value). How strong would an unmeasured confounder have to be to explain an observed risk ratio $RR > 1$ away? VanderWeele & Ding (2017): its risk-ratio associations with both treatment and outcome must be at least

$$E = RR + \sqrt{RR\,(RR-1)} \qquad (\text{use } 1/RR \text{ if } RR < 1). \tag{14.7}$$

For the kidney data the adjusted risk ratio is $RR = 0.83255/0.77887 = 1.0689$ (unrounded values of Worked Example 14.2), so $E = 1.0689 + \sqrt{1.0689 \times 0.0689} = 1.0689 + \sqrt{0.0737} = 1.0689 + 0.2714 = 1.340$: a modest confounder (for instance a second severity dimension not captured by stone size) could erase the advantage of A. For comparison, $RR = 1.5$ gives $E = 1.5 + \sqrt{0.75} = 2.366$ and $RR = 1.8$ gives $E = 1.8 + \sqrt{1.8 \times 0.8} = 1.8 + 1.2 = 3.000$.

PYTHON LAB · s14_examples.py · ex_evalue()

```
kidney adjusted risk ratio RR = 0.83255/0.77887 = 1.0689; E-value = 1.0689 +
0.2714 = 1.340
RR = 1.5 -> E-value = 2.366; RR = 1.8 -> E-value = 3.000
```

PART IV

From Identified Functionals to Estimators

Consistency, the delta method, the bootstrap, instruments and bounds

PART IV · MATHEMATICAL STATISTICS

14.11 The Plug-in Back-door Estimator

Identification delivers a *functional* $\tau(P)$ of the observational law; estimation replaces P by the empirical distribution. For a binary treatment, binary outcome and discrete back-door set Z write $p_z = P(Z{=}z)$, $e_z = P(X{=}1 \mid z)$, $\mu_{xz} = P(Y{=}1 \mid x, z)$, $\tau_z = \mu_{1z} - \mu_{0z}$ and

$$\tau = \sum_z p_z\,\tau_z, \qquad \hat\tau = \sum_z \hat p_z(\hat\mu_{1z} - \hat\mu_{0z}), \qquad \hat p_z = \frac{N_z}{N}, \quad \hat\mu_{xz} = \frac{1}{N_{xz}} \sum_{i:\,Z_i=z, X_i=x} Y_i,$$

where N is the sample size, N_z the number of units with $Z_i = z$ and N_{xz} the number with $(X_i, Z_i) = (x, z)$; $O_i = (Z_i, X_i, Y_i)$ denotes the i-th observation.

Theorem 14.6 (Consistency and asymptotic normality of the plug-in estimator)**.** *Let (Z_i, X_i, Y_i) be i.i.d., Z take finitely many values with $p_z > 0$, and assume positivity $0 < e_z < 1$. Then $\hat\tau \xrightarrow{p} \tau$ and*

$$\sqrt{N}(\hat\tau - \tau) \xrightarrow{d} N(0, V), \qquad V = \underbrace{\sum_z p_z \Big[\frac{\mu_{1z}(1-\mu_{1z})}{e_z} + \frac{\mu_{0z}(1-\mu_{0z})}{1-e_z}\Big]}_{\text{within strata}} + \underbrace{\sum_z p_z(\tau_z - \tau)^2}_{\text{between strata}}.$$

Proof. Consistency. $N_z/N \to p_z$ and $N_{xz}/N \to p_z e_z^x (1-e_z)^{1-x} > 0$ almost surely (strong law), so $\hat\mu_{xz} \to \mu_{xz}$ a.s.; $\hat\tau$ is a continuous function of these finitely many averages, and the continuous

mapping theorem gives $\hat\tau \to \tau$.

Linearization. Since $\sum_z (\hat p_z - p_z) = 0$, adding and subtracting $\sum_z \hat p_z \tau_z$ gives

$$\hat\tau - \tau = \sum_z (\hat p_z - p_z)(\tau_z - \tau) + \sum_z \hat p_z \big[(\hat\mu_{1z} - \mu_{1z}) - (\hat\mu_{0z} - \mu_{0z})\big].$$

The first sum equals $N^{-1}\sum_i (\tau_{Z_i} - \tau)$, because $\hat p_z = N^{-1}\sum_i \mathbb{1}(Z_i{=}z)$ and $\sum_z p_z(\tau_z - \tau) = 0$. Write $\hat p_z(\hat\mu_{1z} - \mu_{1z}) = \frac{N_z}{N_{1z}} \cdot \frac{1}{N}\sum_i \mathbb{1}(Z_i{=}z, X_i{=}1)(Y_i - \mu_{1z})$. The ratio $N_z/N_{1z} \to 1/e_z$ a.s., and $N^{-1/2}\sum_i \mathbb{1}(Z_i{=}z, X_i{=}1)(Y_i - \mu_{1z}) = O_p(1)$ by the CLT; by Slutsky, replacing N_z/N_{1z} by $1/e_z$ changes $\sqrt{N}(\hat\tau - \tau)$ by $o_p(1)$. The same holds for the control arm. Therefore $\sqrt{N}(\hat\tau - \tau) = N^{-1/2}\sum_i \varphi(O_i) + o_p(1)$ with the *influence function*

$$\varphi(O) = \tau_Z - \tau + \frac{\mathbb{1}(X{=}1)(Y - \mu_{1Z})}{e_Z} - \frac{\mathbb{1}(X{=}0)(Y - \mu_{0Z})}{1 - e_Z}.$$

CLT. $\mathbb{E}\varphi = 0$ because $\mathbb{E}[\tau_Z] = \tau$ and $\mathbb{E}[Y - \mu_{XZ} \mid X, Z] = 0$. The three terms are uncorrelated: the last two have conditional mean zero given (X, Z), while the first is a function of Z, and the product of the last two vanishes because $\mathbb{1}(X{=}1)\mathbb{1}(X{=}0) = 0$. Their variances are $\sum_z p_z(\tau_z - \tau)^2$, $\mathbb{E}\big[e_Z\mu_{1Z}(1 - \mu_{1Z})/e_Z^2\big] = \sum_z p_z \mu_{1z}(1 - \mu_{1z})/e_z$ and its control analogue. The classical CLT for the i.i.d. $\varphi(O_i)$ and Slutsky's lemma finish the proof. □

MATHEMATICAL STATISTICS · The delta method and the influence function

Theorem 14.6 is the delta method for $\hat\tau = g(\hat{\boldsymbol\theta})$, a smooth function of cell frequencies, organised through the influence function: $V = \mathrm{Var}\varphi(O)$, estimated by plugging in $\hat p_z, \hat e_z, \hat\mu_{xz}$. The residual terms of φ are the IPW correction of Proposition 14.2; φ is the *efficient* influence function of the ATE (Session 15), so the saturated plug-in estimator is efficient for discrete Z. Positivity enters through $1/e_z$ and $1/(1 - e_z)$. The nonparametric bootstrap is valid for the same smoothness reason (Wasserman 2004, Ch. 8) and fails near positivity violations, where a resample may lack treated units in a stratum.

Worked Example 14.15 (A 95% confidence interval for the kidney effect)**.** *(Numerical, data of Table 14.2, treatment $X = 1$ for A.)* The stratum effects are $\hat\tau_s = 0.9310 - 0.8667 = 0.0644$ (small) and $\hat\tau_\ell = 0.7300 - 0.6875 = 0.0425$ (large), so $\hat\tau = 0.51(0.0644) + 0.49(0.0425) = 0.0537$. With $\hat e_s = 0.2437$, $\hat e_\ell = 0.7668$ the within-strata terms of Theorem 14.6 are

$$\text{small:}\quad 0.51\left[\frac{0.9310(0.0690)}{0.2437} + \frac{0.8667(0.1333)}{0.7563}\right] = 0.51\,(0.2635 + 0.1528) = 0.2123,$$

$$\text{large:}\quad 0.49\left[\frac{0.7300(0.2700)}{0.7668} + \frac{0.6875(0.3125)}{0.2332}\right] = 0.49\,(0.2570 + 0.9211) = 0.5773,$$

summing to 0.7896; the between-strata term is only $0.51(0.0644 - 0.0537)^2 + 0.49(0.0425 - 0.0537)^2 = 0.00012$. Hence $\hat V = 0.7897$, $\widehat{\mathrm{SE}} = \sqrt{0.7897/700} = 0.0336$ and the 95% delta-method interval is $0.0537 \pm 1.96(0.0336) = 0.0537 \pm 0.0658 = [-0.0122, 0.1195]$ (computed from unrounded values): it contains 0. A nonparametric bootstrap ($B = 4000$ resamples of the 700 patients with replacement, i.e. multinomial draws over the eight (size, treatment, outcome) cells; seed 9) gives $\mathrm{SE} = 0.0335$ and the percentile interval $[-0.0125, 0.1229]$. The real data establish that the naive comparison is *wrong*; they do not establish, at the 5% level, that A is better.

PYTHON LAB · s14_examples.py · ex_delta_ci_kidney()

```
tau_hat = 0.51(0.0644) + 0.49(0.0425) = 0.0537; within:  small 0.2123, large
0.5773
V = within 0.7896 + between 0.00012 = 0.7897; SE = sqrt(V/700) = 0.0336;
1.96 x SE = 0.0658
95% delta-method CI: [-0.0122, 0.1195]; bootstrap (B=4000, seed=9):  SE =
0.0335, CI = [-0.0125, 0.1229]
```

PART IV · MATHEMATICAL STATISTICS

14.12 Instrumental-Variable Estimation

Proposition 14.4 (Consistency of the IV estimator). *In the model of Proposition 14.3, the sample IV estimator* $\hat\beta_{IV} = \widehat{\mathrm{Cov}}(Z,Y)/\widehat{\mathrm{Cov}}(Z,X)$ *satisfies* $\hat\beta_{IV} \xrightarrow{p} \beta$. *If instead* Z *has a direct effect* θ *on* Y *(exclusion fails,* $Y = \beta X + \gamma U + \theta Z + \varepsilon_Y$*),* $\hat\beta_{IV} \xrightarrow{p} \beta + \theta/\alpha$.

Proof. Sample covariances converge in probability to population covariances (weak law of large numbers for $\overline{ZY}, \bar Z, \bar Y$ and continuous mapping). Since $\mathrm{Cov}(Z,X) = \alpha \mathrm{Var} Z \neq 0$, the ratio map is continuous at the limit, and Proposition 14.3 gives β. With the extra term θZ in Y, $\mathrm{Cov}(Z,Y) = \beta\alpha \mathrm{Var} Z + \theta \mathrm{Var} Z$, so the ratio tends to $(\beta\alpha \mathrm{Var} Z + \theta \mathrm{Var} Z)/(\alpha \mathrm{Var} Z) = \beta + \theta/\alpha$. □

The second statement is a warning: a small exclusion violation θ is divided by the first-stage strength α, so weak instruments amplify every flaw.

MATHEMATICAL STATISTICS · Weak instruments

Consistency is a large-N statement. When α is small, the denominator $\widehat{\mathrm{Cov}}(Z,X)$ is near zero: $\hat\beta_{IV}$ is heavy-tailed (no finite moments in the just-identified Gaussian case) and median-biased towards OLS. The standard diagnostic is the first-stage F statistic — the squared t-statistic of the coefficient of Z in the regression of X on Z — with the rule of thumb $F > 10$ (Staiger & Stock 1997); its mean is roughly $1 + n\alpha^2 \mathrm{Var} Z / \mathrm{Var}(U + \varepsilon_X)$. *Simulation* (`ex_iv_sim()`, model of Worked Example 14.11, 2000 samples of $n = 500$ each, same random stream continued from seed 14; Fig. 14.7, right): $\alpha = 0.8$ gives mean $F = 41.1$ (approximation $1 + 500(0.64)(0.25)/2 = 41.0$), a median IV estimate 1.007 and interquartile range (IQR) $[0.831, 1.161]$, of width 0.330; $\alpha = 0.1$ gives $F = 1.7$ (approximation 1.62), median 1.548 and IQR $[0.528, 2.409]$, of width 1.882, i.e. 5.7 times wider. The weak-instrument median has moved $(1.548 - 1)/(1.926 - 1) = 0.59$ of the way from $\beta = 1$ to the OLS limit 1.926.

PYTHON LAB · s14_examples.py · ex_iv_sim()

```
alpha=0.8:  n=500, 2000 reps:  median IV = 1.007, IQR = [0.831, 1.161]
(width 0.330), mean F = 41.1
alpha=0.1:  n=500, 2000 reps:  median IV = 1.548, IQR = [0.528, 2.409]
(width 1.882), mean F = 1.7
share of the way to the OLS limit:  (1.548 - 1)/(1.926 - 1) = 0.59; IQR
ratio 1.882/0.330 = 5.7
```

PART IV · MATHEMATICAL STATISTICS

14.13 Partial Identification

Proposition 14.5 (No-assumption bounds; Manski 1990)**.** *For binary X, Y and* any *causal structure, let $Y(x)$ be the potential outcome (Session 13: the value of Y in the submodel M_x), so that $P(Y{=}1 \mid \mathrm{do}(x)) = P(Y(x){=}1)$. Then*

$$P(X{=}x, Y{=}1) \;\le\; P(Y{=}1 \mid \mathrm{do}(x)) \;\le\; P(X{=}x, Y{=}1) + P(X \neq x),$$

and the interval for the ATE always has width exactly 1*.*

Proof. By consistency $Y = Y(x)$ when $X = x$, so $P(Y(x){=}1) = P(X{=}x, Y{=}1) + P(X{\neq}x,\, Y(x){=}1)$. The last probability is not constrained by the data and can be anything in $[0, P(X \neq x)]$, giving the bounds; both ends are attained by SCMs compatible with the data. For the ATE, subtracting the bounds for $x = 0$ from those for $x = 1$ gives an interval of width $P(X{\neq}1) + P(X{\neq}0) = 1$. □

MATHEMATICAL STATISTICS · What bounds buy

In Worked Example 14.14, $P(X{=}1, Y{=}1) = 0.40$, $P(X{=}0, Y{=}1) = 0.15$ and $P(X{=}1) = P(X{=}0) = 0.5$, so $P(Y{=}1 \mid \mathrm{do}(1)) \in [0.40,\, 0.40 + 0.50] = [0.40, 0.90]$ and $P(Y{=}1 \mid \mathrm{do}(0)) \in [0.15,\, 0.15{+}0.50] = [0.15, 0.65]$. The ATE bounds are $[0.40{-}0.65,\, 0.90{-}0.15] = [-0.25, 0.75]$, of width 1: the data cannot sign the effect but rule out values below -0.25. Each added assumption narrows the interval (monotone treatment response $Y(1) \ge Y(0)$ lifts the lower bound to 0; an instrument gives the Balke–Pearl bounds), making the price of every assumption visible.

PART IV · MATHEMATICAL STATISTICS

14.14 Worked Examples

Worked Example 14.16 (An online-experimentation graph, resolving Session 12's ad vignette)**.** Recall Session 12's marketing example: ad exposure X, conversion Y, confounded by purchase intent Z (which drives both the targeting algorithm's choice to show the ad and the baseline conversion probability). Z satisfies the back-door criterion exactly as stone severity did. Using the hypothetical numbers from Session 12 (25% conversion for high-intent customers regardless of ad, 5% for low-intent, 70% of ad-shown customers high-intent vs. 20% of not-shown), and assuming in addition (Session 12 does not specify it) an overall intent prevalence $P(Z{=}\text{high}) = 0.4$ across the full customer base. By the law of total probability $P(Z{=}\text{high}) = 0.7p + 0.2(1 - p)$ with $p = P(X{=}1)$, so $0.4 = 0.2 + 0.5p$ and the ad was shown to $p = 0.40$ of customers. The naive rates are $P(Y{=}1 \mid X{=}1) = 0.7(0.25) + 0.3(0.05) = 0.19$ and $P(Y{=}1 \mid X{=}0) = 0.2(0.25) + 0.8(0.05) = 0.09$. Back-door adjustment instead weights by $P(z)$: $P(Y{=}1 \mid \mathrm{do}(X{=}1)) = 0.25(0.4) + 0.05(0.6) = 0.13$, and identically $P(Y{=}1 \mid \mathrm{do}(X{=}0)) = 0.25(0.4) + 0.05(0.6) = 0.13$ — the back-door-adjusted effect of the ad is exactly zero, correctly recovering the fact (stated by construction in Session 12) that the ad had no true causal effect at all, in sharp contrast to the naive 19% vs. 9% comparison that

ignored intent entirely.

PYTHON LAB · s14_examples.py · ex_ad_backdoor()

```
naive P(Y=1|X=1) = 0.19, P(Y=1|X=0) = 0.09; P(Y=1|do(X=1)) = 0.13,
P(Y=1|do(X=0)) = 0.13
consistency:  0.7 p + 0.2 (1-p) = 0.4 => P(X=1) = p = 0.2/0.5 = 0.40
```

Additional Worked Examples (Instructor Supplement)

Worked Example 14.17 (Back-door adjustment with a genuinely two-dimensional adjustment set)**.** Every numerical back-door example above used a single confounder Z. Let $X \to Y$ be confounded by *two* independent variables: $Z_1 \to X$, $Z_1 \to Y$, $Z_2 \to X$, $Z_2 \to Y$, with $P(Z_1{=}1) = 0.4$, $P(Z_2{=}1) = 0.6$ (independent), and $P(Y{=}1 \mid X{=}1, z_1, z_2) = 0.5 + 0.1z_1 + 0.1z_2$, $P(Y{=}1 \mid X{=}0, z_1, z_2) = 0.3 + 0.1z_1 + 0.1z_2$ (so the true effect of X is a constant 0.2 regardless of z_1, z_2). Confirm that $\{Z_1, Z_2\}$ recovers this exactly.
Solution. By independence, $P(z_1, z_2)$ for the four combinations is $P(0,0){=}0.6(0.4){=}0.24$, $P(1,0){=}0.4(0.4){=}0.16$, $P(0,1){=}0.6(0.6){=}0.36$, $P(1,1){=}0.4(0.6){=}0.24$ (summing to 1). Then

$$\begin{aligned} P(Y{=}1 \mid \mathrm{do}(X{=}1)) &= 0.5(0.24) + 0.6(0.16) + 0.6(0.36) + 0.7(0.24) \\ &= 0.12 + 0.096 + 0.216 + 0.168 = \mathbf{0.60}, \\ P(Y{=}1 \mid \mathrm{do}(X{=}0)) &= 0.3(0.24) + 0.4(0.16) + 0.4(0.36) + 0.5(0.24) \\ &= 0.072 + 0.064 + 0.144 + 0.12 = \mathbf{0.40}. \end{aligned}$$

The recovered effect $0.60 - 0.40 = \mathbf{0.20}$ matches the true, constant structural effect exactly. The back-door adjustment formula $P(y \mid \mathrm{do}(x)) = \sum_{z_1,z_2} P(y \mid x, z_1, z_2) P(z_1, z_2)$ generalizes immediately to a vector-valued Z; the only practical change is that the sum now ranges over every joint configuration of all adjustment variables, which is exactly why Session 15 will need machine learning (rather than a hand-built table) once Z has dozens of real-valued covariates instead of two binary ones.

PYTHON LAB · s14_examples.py · ex_two_var_adjustment()

```
P(Y=1|do(X=1)) = 0.60, P(Y=1|do(X=0)) = 0.40, effect = 0.20
```

Worked Example 14.18 (A valid back-door set need not be minimal)**.** Consider $Z_2 \to Z_1$, $Z_1 \to X$, $Z_1 \to Y$, $X \to Y$ (Z_2 is an ancestor of Z_1 but has no other connection to X or Y). Does $\{Z_1\}$ satisfy the back-door criterion? Does $\{Z_1, Z_2\}$?
Solution. $\{Z_1\}$: not a descendant of X, and it blocks the only back-door path $X \leftarrow Z_1 \to Y$ (a fork), so $\{Z_1\}$ is valid — and minimal. $\{Z_1, Z_2\}$: Z_2 is also not a descendant of X (it is further upstream), and the back-door path is still blocked by $Z_1 \in \{Z_1, Z_2\}$ regardless of whether Z_2 is additionally included, so $\{Z_1, Z_2\}$ is *also* a valid back-door set — just not a minimal one. Including a harmless ancestor of an already-sufficient adjustment variable never breaks validity; it can only cost estimation efficiency later (Session 15: more covariates to adjust for means more parameters to estimate from finite data), never correctness. This is the opposite lesson from the set $\{Z_1, Z_3, W\}$ of Worked Example 14.4 (which included a genuine *descendant* of X and thereby broke condition (i)) and from $\{Z_3\}$ there (which opened a collider) — the

back-door criterion is sensitive to which *kind* of extra variable you add, not merely to how many you add.

PART V

Identification in Agents, RL and Multi-Agent Systems

Every action is an intervention; every log is observational data

PART V · AI CONNECTIONS

14.15 Reinforcement Learning: Actions Are Interventions

An RL agent does not observe its actions — it *sets* them. With $G_t = \sum_{k\geq 0} \gamma^k R_{t+k+1}$ the discounted return ($0 \leq \gamma < 1$), the action value is an interventional quantity,

$$Q^\pi(s,a) = \mathbb{E}\big[G_t \mid S_t = s,\ \mathrm{do}(A_t = a),\ \text{then follow } \pi\big], \qquad V^\pi = \mathbb{E}\big[G_t \mid \mathrm{do}(\pi)\big],$$

and the value of a policy is exactly the policy intervention of Proposition 14.1. Online, the agent's own randomization makes conditioning and doing coincide. Offline, the data come from a behaviour (logging) policy $\mu(a \mid s)$, and identification is required; estimating a target policy's value from such logs is *off-policy evaluation* (OPE): if μ chose actions using only the recorded state, S is a back-door set for (A, R) and both the adjustment ("direct method") and the IPW ("importance sampling") forms of Proposition 14.2 apply (Fig. 14.9).

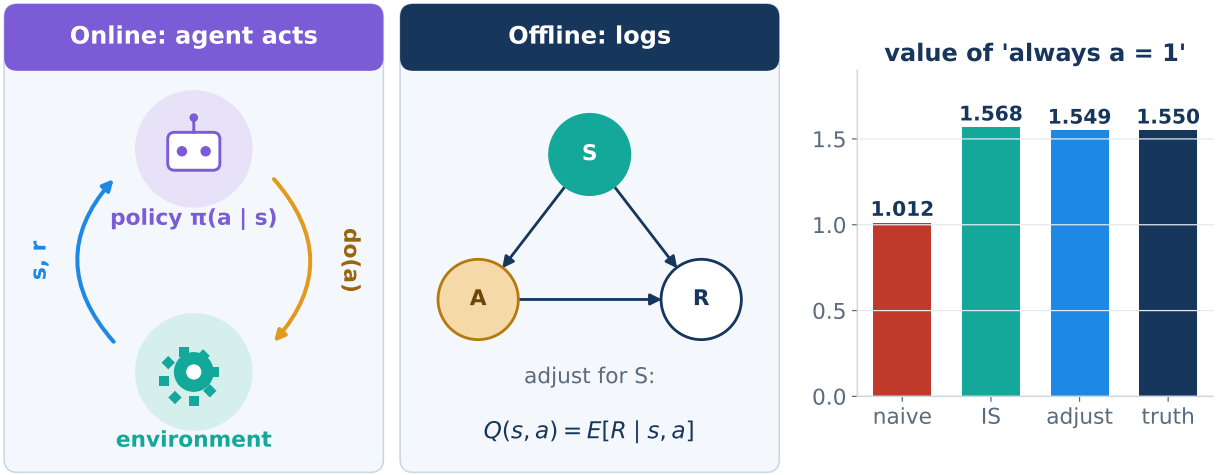


Figure 14.9. Left: online, the agent intervenes. Middle: offline logs form the back-door graph $A \leftarrow S \rightarrow R$. Right: the value of "always $a = 1$" in the logged bandit of Worked Example 14.19: naive conditioning is badly biased; importance sampling and adjustment recover the truth.

Worked Example 14.19 (Off-policy evaluation as identification)**.** *(Numerical, hypothetical bandit.)* Context $S \in \{0,1\}$ with $P(S{=}1) = 0.3$; the logging policy plays $A = 1$ with probability $\mu(1 \mid 0) = 0.2$ in $s = 0$ and $\mu(1 \mid 1) = 0.9$ in $s = 1$. Mean rewards $r(s,a) = \mathbb{E}[R \mid s, a]$: $r(0,0) = 1$, $r(0,1) = 2$, $r(1,0) = 0$, $r(1,1) = 0.5$. The target policy "always $a = 1$" has value $\sum_s P(s) r(s,1) = 0.7(2) + 0.3(0.5) = 1.4 + 0.15 = 1.55$. The naive logged average weights the

contexts by $P(s \mid A{=}1)$: $P(S{=}0, A{=}1) = 0.7(0.2) = 0.14$ and $P(S{=}1, A{=}1) = 0.3(0.9) = 0.27$, so $\mathbb{E}[R \mid A{=}1] = (0.14(2) + 0.27(0.5))/0.41 = 0.415/0.41 = 1.012$, dominated by the poor context $s = 1$, where $a = 1$ was logged most. *Simulation* (`ex_rl_ope()`): $n = 20\,000$ logged rounds, rewards $r(s, a)$ plus Gaussian noise with sd 0.5, seed 17. Importance sampling $\frac{1}{n}\sum_i \mathbb{1}(A_i{=}1) R_i / \mu(1 \mid S_i) = 1.568$; adjustment ("direct method") $\sum_s \hat{P}(s)\hat{Q}(s, 1) = 1.549$, where $\hat{P}(s)$ is the logged frequency of context s and $\hat{Q}(s, 1)$ the average logged reward over rounds with $S = s$, $A = 1$.

PYTHON LAB · `s14_examples.py` · `ex_rl_ope()`

```
P(S=0,A=1) = 0.14, P(S=1,A=1) = 0.27; naive = 0.415/0.41
true V(always a=1) = 1.550; naive E[R|A=1] = 1.012
logged n=20000, seed=17:  importance sampling = 1.568; direct (adjustment) =
1.549
```

LINK TO COURSE SESSION 17 · RL and MDPs

The Bellman equation $Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^\pi(s')$ silently uses $P(s' \mid s, a) = P(s' \mid s, \mathrm{do}(a))$: transitions are mechanisms, and the Markov state is a back-door set for every action. When the state is only partially observed (partially observable MDPs, POMDPs, course session 21), this fails and identification must be argued explicitly.

LINK TO COURSE SESSION 20 · Deep and Model-based RL

Off-policy evaluation and offline RL are identification problems first: the weights $\pi(a \mid s)/\mu(a \mid s)$ are the IPW form of Proposition 14.2 and need positivity and a logger that used no unrecorded information. Policy gradients (course session 19) differentiate exactly the policy value Eq. (14.2).

PART V · AI CONNECTIONS

14.16 Agentic AI: What If the Agent Always Called the Tool?

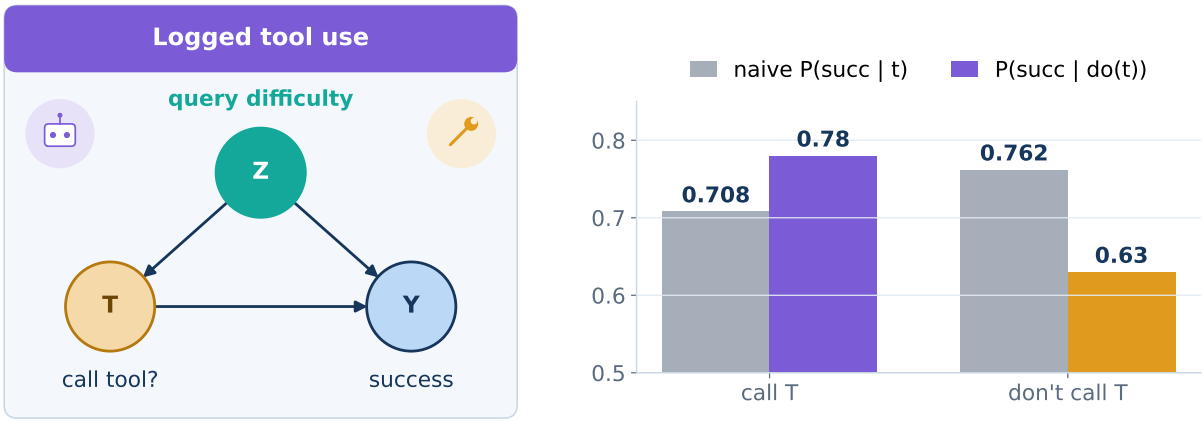


Figure 14.10. Logged tool use of an LLM agent (hypothetical, Worked Example 14.20). Query difficulty confounds the decision to call tool T and task success; the naive comparison has the wrong sign.

Worked Example 14.20 (Can the logs answer the product question?)**.** *(Numerical, hypothetical.)* An agent's past policy called a retrieval tool T on 80% of hard queries and 30% of easy ones; 40% of queries are hard (query difficulty Z). Success rates: hard 0.6 with T, 0.3 without; easy 0.9 with T, 0.85 without. The joint frequencies are $P(T, \text{hard}) = 0.4(0.8) = 0.32$, $P(T, \text{easy}) = 0.6(0.3) = 0.18$, $P(\text{no } T, \text{hard}) = 0.08$, $P(\text{no } T, \text{easy}) = 0.42$, so half of all logged queries used T. The naive success rates are $P(\text{succ} \mid T) = (0.32(0.6) + 0.18(0.9))/0.5 = 0.354/0.5 = 0.708$ and $P(\text{succ} \mid \text{no } T) = (0.08(0.3) + 0.42(0.85))/0.5 = 0.381/0.5 = 0.762$: "the tool hurts" ($-0.054$). Difficulty is recorded and is the only common cause, so it is a back-door set: $P(\text{succ} \mid \text{do}(T)) = 0.4(0.6) + 0.6(0.9) = 0.24 + 0.54 = 0.78$ and $P(\text{succ} \mid \text{do}(\text{no } T)) = 0.4(0.3) + 0.6(0.85) = 0.12 + 0.51 = 0.63$, an effect of $+0.15$. The conditional policy "call T only on hard queries" (Eq. (14.2)) reaches $0.4(0.6) + 0.6(0.85) = 0.75$ with only 0.40 tool calls per query — a cost–benefit trade-off computed entirely from logs.

PYTHON LAB · `s14_examples.py` · `ex_agent_tool()`

```
P(T) = 0.50; numerators:  T 0.192 + 0.162 = 0.354, noT 0.024 + 0.357 = 0.381
naive:  P(succ|T) = 0.708, P(succ|noT) = 0.762, diff = -0.054
adjusted:  P(succ|do(T)) = 0.78, P(succ|do(noT)) = 0.63, effect = +0.15;
policy 'T only on hard' 0.75
```

LINK TO COURSE SESSION 26 · Agentic AI

Agents that learn when to call tools, retrieve or reflect from their own traces face exactly this question. Run the recipe of Section 14.10 first: what did the old policy see when it decided? If an unlogged signal (user expertise, an internal confidence score) drove both decision and outcome, the effect may be unidentified; an online A/B test or a front-door variable (was the tool output actually used?) is then needed. Planners (course sessions 23–24) that simulate $P(s' \mid \text{do}(a))$ with a model fitted to confounded traces inherit the same bias.

Table 14.4. This session in the Advanced AI course.

Concept today	Where it reappears	Course session
$\text{do}(x)$, $\text{do}(\pi)$	action values, policy values, policy gradients	17, 18, 19
Back-door / IPW	importance-sampling OPE, offline RL, logged tool use	15, 20, 26
Instruments	randomized encouragement, exploration noise as an instrument	18, 26
Non-identifiability, bounds	POMDPs with hidden confounders, safe decisions under ambiguity	21, 31
Total vs. direct effects	mechanism design, credit assignment	27

PART V · AI CONNECTIONS

14.17 Multi-Agent AI: Interventions Ripple Through Other Agents

Intervening on one agent changes the inputs of the others, who may adapt; "the effect of agent 1" is ambiguous until we say what happens to everyone else.

Worked Example 14.21 (Total versus controlled direct effect of an agent)**.** *(Numerical, hypothetical.)* Agent 1 chooses $X \in \{0,1\}$; agent 2 observes it and plays $A_2 = 1$ with probability 0.7 if $X = 1$ and 0.2 if $X = 0$; the team reward has $\mathbb{E}[Y \mid x, a_2] = 0.2, 0.5, 0.4, 0.9$ for $(x, a_2) = (0,0), (0,1), (1,0), (1,1)$. Nothing else influences X, so $\mathbb{E}[Y \mid \mathrm{do}(x)] = \sum_{a_2} P(a_2 \mid x)\mathbb{E}[Y \mid x, a_2]$. The *total* effect with agent 2 adapting is therefore $\mathbb{E}[Y \mid \mathrm{do}(X{=}1)] - \mathbb{E}[Y \mid \mathrm{do}(X{=}0)] = \big(0.3(0.4) + 0.7(0.9)\big) - \big(0.8(0.2) + 0.2(0.5)\big) = (0.12 + 0.63) - (0.16 + 0.10) = 0.75 - 0.26 = 0.49$. Freezing agent 2 by a second intervention gives *controlled direct* effects $\mathbb{E}[Y \mid x{=}1, a_2] - \mathbb{E}[Y \mid x{=}0, a_2]$: $0.4 - 0.2 = 0.20$ under $\mathrm{do}(A_2{=}0)$ and $0.9 - 0.5 = 0.40$ under $\mathrm{do}(A_2{=}1)$. Three legitimate numbers answer three different questions; identifying the direct effects from logs requires that nothing unobserved confounds A_2 and Y given X — in a game, shared private signals often do.

PYTHON LAB · s14_examples.py · ex_multiagent()

```
E[Y|do(X=1)] = 0.75, E[Y|do(X=0)] = 0.26, total effect = 0.49
controlled direct effect with do(A2=0):  0.20; with do(A2=1):  0.40
```

LINK TO COURSE SESSION 27 · Multi-Agent AI

Mechanism design intervenes on rules every agent reacts to: its effect is a total effect. Counterfactual credit assignment (counterfactual multi-agent policy gradients, COMA, Session 13) uses controlled direct effects. In logs of agent populations, matchmaking, reputation and shared information are confounders to block or randomize away.

AI CONNECTION · Generative Models and AI Paradigms

A deep generative model fitted to observational data (course session 4) learns $P(\mathbf{v})$ — rung 1 of the ladder of causation (Session 12); even a perfect VAE or diffusion model cannot answer $P(y \mid \mathrm{do}(x))$ in a non-identified graph (Theorem 14.5 holds for any model class). Do-calculus is symbolic inference over graphs (course session 2); pairing it with neural estimators is neuro-symbolic AI in miniature: symbols decide *what* to estimate, networks estimate it.

PART V · AI CONNECTIONS

14.18 Summary

THINK ABOUT IT **A hospital argues that because its database has two million patients, the effect of a drug is "certainly identified". What is wrong with this argument, and what would you ask for instead?**

THINK ABOUT IT **In Worked Example 14.4, $\{Z_1, Z_3\}$ and $\{Z_2, Z_3\}$ are both valid. If Z_1 is cheap to measure but noisy and Z_2 expensive but precise, which would you choose, and why might the answer differ for bias and for variance?**

THINK ABOUT IT **An LLM agent's logs contain the model's internal confidence score only for runs after a software update. Which identification strategies remain available for runs before the update?**

Table 14.5. Cheat sheet of key formulas.

Object	Formula / condition
Truncated factorization	$P(\mathbf{v} \mid \mathrm{do}(x)) = \prod_{i \notin X} P(v_i \mid \mathrm{pa}_i)\big\vert_{X=x}$
Back-door criterion	$Z \cap \mathrm{de}(X) = \emptyset$ and $X \perp\!\!\!\perp_d Y \mid Z$ in $G_{\underline{X}}$
Back-door adjustment	$P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z) = \mathbb{E}\big[\mathbb{1}(X{=}x)\mathbb{1}(Y{=}y)/P(x \mid Z)\big]$
Policy value	$P(y \mid \mathrm{do}(\pi)) = \sum_{z,x} P(y \mid x, z)\pi(x \mid z)P(z)$
Front-door	$P(y \mid \mathrm{do}(x)) = \sum_m P(m \mid x) \sum_{x'} P(y \mid x', m)P(x')$
Do-calculus	Rule 1: $G_{\overline{X}}$; Rule 2: $G_{\overline{X}\underline{Z}}$; Rule 3: $G_{\overline{X}\,\overline{Z(W)}}$
IV (linear)	$\beta = \mathrm{Cov}(Z, Y)/\mathrm{Cov}(Z, X)$; exclusion failure adds θ/α
Plug-in variance	$V = \sum_z p_z \big[\frac{\mu_{1z}(1-\mu_{1z})}{e_z} + \frac{\mu_{0z}(1-\mu_{0z})}{1-e_z} + (\tau_z - \tau)^2\big]$
Bounds; E-value	$P(X{=}x, Y{=}1) \le P(Y{=}1 \mid \mathrm{do}(x)) \le P(X{=}x, Y{=}1) + P(X{\neq}x)$; $E = RR + \sqrt{RR(RR-1)}$

Table 14.6. Common pitfalls.

Pitfall	Remedy
"More data will identify it"	identification is about P, not n; check the graph first
Adjusting for all pre-treatment variables	pre-treatment colliders (Z_3 in Worked Example 14.4: M-bias) open paths; use the criterion
Adjusting for a mediator or any descendant of X	condition (i); the effect through it is lost (Worked Example 14.6)
Using front-door when U touches M	condition (ii) fails; the effect is then typically unidentified
Reporting a point estimate for an unidentified effect	report bounds and sensitivity (E-value) instead

Further reading. Pearl (2009), *Causality*, Ch. 3–4; Pearl, Glymour & Jewell (2016), *Causal Inference in Statistics: A Primer*, Ch. 3; Hernán & Robins (2020), *Causal Inference: What If*, Ch. 7, 12, 16; Peters, Janzing & Schölkopf (2017), Ch. 6; Shpitser & Pearl (2006) on the ID algorithm; Wasserman (2004), *All of Statistics*, Ch. 8.

APPENDIX

14.19 Appendix: Back-door Adjustment from Do-Calculus

We derive Theorem 14.1 from Theorem 14.3 for Z satisfying the back-door criterion. Start from $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z)P(z \mid \mathrm{do}(x))$ (law of total probability in the post-intervention model).

Step 1 (Rule 3). Since Z contains no descendants of X, intervening on X cannot affect Z's distribution, so $P(z \mid \mathrm{do}(x)) = P(z)$. Formally, Rule 3 (with no outer intervention and $W = \emptyset$, so $Z(W) = X$) requires $Z \perp\!\!\!\perp_d X$ in $G_{\overline{X}}$. In $G_{\overline{X}}$ every path between X and a node of Z leaves X along an outgoing arrow $X \to \cdots$. Because no node of Z is a descendant of X, the path cannot stay directed all the way, so it must change direction somewhere; the first change of direction from $\to$ to $\leftarrow$ is a collider. Nothing is conditioned on, so that collider blocks the path.

Step 2 (Rule 2). Since Z blocks every back-door path, $Y \perp\!\!\!\perp_d X \mid Z$ holds in $G_{\underline{X}}$ (removing X's outgoing edges leaves only back-door paths, all blocked by Z), so by Rule 2, $P(y \mid \mathrm{do}(x), z) = P(y \mid x, z)$.

Substituting both simplifications gives $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z)$, exactly Theorem 14.1. The same two steps with $\pi(x \mid z)$ in place of a point mass give Proposition 14.1.

CONNECTIONS AND TAKEAWAYS

We can now determine, directly from a graph, *whether* $P(y \mid \mathrm{do}(x))$ is a well-defined function of observational data (back-door, front-door, instruments with extra assumptions, or in general via do-calculus and the ID algorithm) and, when it is, write that function down explicitly; when it is not, we can bound it and measure how sensitive a conclusion is to hidden confounding. This is a complete answer to the identification question posed at the top of this session.

What we have *not* done systematically is estimate these quantities from finite, noisy, high-dimensional data — apart from the plug-in, bootstrap, IV and importance-sampling calculations of Parts IV and V, every formula above was applied either to population-level quantities or to a dataset treated as if it perfectly revealed $P(y \mid x, z)$ and $P(z)$. **Session 15**, the final session of these notes, closes this last gap: given real observational data, how do we actually build good estimators of the identified causal effect — propensity scores, inverse propensity weighting, doubly robust estimation, and machine-learning-based treatment-effect estimators — and how does the special case of a randomized experiment (where identification is guaranteed by design) connect back to everything built since Session 12?